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Glycolysis-dependent Sulfur Metabolism Orchestrates Morphological Plasticity and Virulence in Fungi

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eLife Assessment

This work identifies a novel, conserved link between glycolysis and sulfur metabolism that governs fungal morphogenesis and virulence. The **compelling** evidence, integrating multiple approaches, provides an **important** conceptual advance. A future mechanistic dissection of how sulfur metabolites interface with known pathways is encouraged.

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Abstract

Fungi exhibit remarkable morphological plasticity, which allows them to undergo reversible transitions between distinct cellular states in response to changes in their environment. This phenomenon, termed fungal morphogenesis, is critical for fungi to survive and colonize diverse ecological niches and establish infections in a variety of hosts. Despite significant advancements in the field with respect to understanding the gene regulatory networks that control these transitions, the metabolic determinants of fungal morphogenesis remain poorly characterized. In this study, we uncover a previously uncharacterized, conserved dependency between central carbon metabolism and *de novo* biosynthesis of sulfur-containing amino acids that is critical for fungal morphogenesis, in two key fungal species. Using a multidisciplinary approach, we demonstrate that glycolytic flux is crucial to drive fungal morphogenesis in a cAMP-independent manner and perturbation of this pathway leads to a significant downregulation in the expression of genes involved in *de novo* biosynthesis of sulfur-containing amino acids. Remarkably, exogenous supplementation of sulfur-containing amino acids robustly rescues the morphogenesis defect induced by the perturbation of glycolysis in both *Saccharomyces cerevisiae* and *Candida albicans*, underscoring the pivotal role of *de novo* biosynthesis of sulfur-containing amino acid as a downstream effector of morphogenesis. Furthermore, a *C. albicans* mutant lacking the glycolytic enzyme, phosphofructokinase-1 (Pfk1) exhibited significantly reduced survival within *murine* macrophages and attenuated virulence in a *murine* model of systemic candidiasis. Overall, our work elucidates a previously uncharacterized coupling between glycolysis and sulfur metabolism that is critical for driving fungal morphogenesis, contributing to our understanding of this conserved phenomenon.

Introduction

In response to changes in their environment, fungi exhibit remarkable morphological plasticity, allowing them to reversibly transition between specialized cell states. This inherent ability to rapidly adapt to the changes in their environment is essential for their survival and allows them to colonize diverse ecological niches including various host organisms (Gow et al., 2002; Lin et al., 2015; Naranjo-Ortiz and Gabaldón, 2019). This conserved phenomenon, of fungal morphogenesis, encompasses a plethora of morphological forms including pseudohyphae, hyphae, appressoria and fruiting bodies to name a few (Lin et al., 2015; Riquelme et al., 2018). Importantly, these morphological transitions in various pathogenic fungal species significantly influence their interaction with the respective host, impacting their overall ability to cause persistent infections (Min et al., 2020; Wang and Lin, 2012). Fungal morphogenesis is conserved across a plethora of fungal species and has profound implications with respect to fungal ecology, anti-fungal drug resistance and pathogenicity across a spectrum of hosts (Christensen, 1989; Puumala et al., 2024; Sexton and Howlett, 2006). Consequently, it has been a central focus of intense scientific investigation over the last several years (Lin et al., 2015; Riquelme et al., 2018). During dimorphic transitions in fungi, changes in cell shape is accompanied with changes in cell polarity, cell size and cell-cell adhesion and these changes are regulated by two major signalling pathways: the cAMP-PKA pathway and MAP kinase (MAPK) pathways. Both pathways are crucial for morphogenesis in fungal model systems including *Saccharomyces cerevisiae* and *Candida albicans*. In *S. cerevisiae*, the cAMP-PKA pathway is a critical regulatory cascade that controls cellular differentiation, such as pseudohyphal growth, in response to glucose availability. The glucose signal is transduced through two parallel mechanisms: The G-protein coupled receptor (GPCR) Gpr1-Gpa2 system and the intracellular Ras system. The activation of Ras is triggered by high glycolytic flux, which is directly controlled by the Hxt (Hexose Transporter) family dictating the precise rate of glucose import into the cell. Both the Gpr1-Gpa2 and Ras pathways activate the adenylate cyclase (Cyr1), leading to elevated intracellular cAMP levels. This cAMP binds to the regulatory subunit Bcy1, releasing and activating the catalytic Tpk subunits of Protein Kinase A (PKA). This PKA activation, driven primarily by Tpk2, promotes the shift to a filamentous morphology by inducing the expression of necessary genes required for pseudohyphal differentiation (Colombo et al., 1998; Lorenz et al., 2000; Xue et al., 1998). Similarly, this pathway drives the essential yeast to hyphal transition in *C. albicans*, often *via* the regulator Efg1 (Maidan et al., 2005). Concurrently, the MAPK (Mitogen-Activated Protein Kinase) pathway operates in parallel to the cAMP-PKA cascade and coordinates fungal morphogenesis in response to extracellular stress and nutrient signals. In *S. cerevisiae*, the Kss1 cascade is the key effector that drives pseudohyphal growth, predominantly activating under nitrogen-limiting conditions (Lorenz and Heitman, 1998). In the opportunistic fungal pathogen *C. albicans*, the homologous Cek1 pathway is indispensable for coordinating the complex hyphal development. Cek1 acts as a multifaceted sensor, by processing external morphogenetic cues and often engaging directly with the cell wall integrity system to ensure structural stability during the rapid yeast to hyphal transition. This pathway is a critical determinant of virulence, as its activation promotes the expression of adherence factors and ensures the formation of biofilms, enabling the fungus to colonize host tissues and resist immune response (Csank et al., 1998). The cAMP-PKA and MAPK pathways function in a parallel manner, engaging in significant crosstalk to integrate diverse environmental signals and coordinate a precise morphological response (Biswas et al., 2007; Kumar, 2021). While significant progress has been made in elucidating the intricate gene regulatory networks that govern these morphological transitions (e.g., Genetic networks that orchestrate yeast to pseudohyphal differentiation in *S. cerevisiae* and yeast to hyphal differentiation in *C. albicans* have been well characterized (Nantel et al., 2002; Ryan et al., 2012)), a critical knowledge gap persists regarding the underlying metabolic determinants of fungal morphogenesis. Despite the recognized influence of nutrient availability on fungal morphogenesis (Broach, 2012; Fleck et al., 2011; Kumar, 2021), our understanding of the specific metabolic networks and their regulatory influence on morphogenetic switching in fungi remain incompletely characterized.

In this study, we employed *S. cerevisiae* and the human fungal pathogen, *C. albicans* to dissect the metabolic underpinnings of fungal morphogenesis, under nitrogen-limiting conditions, a known trigger for morphogenetic switching in these fungi (Csank et al., 1998; Gimeno et al., 1992). Our findings reveal a critical role for central carbon metabolism, particularly glycolysis, in facilitating the yeast to pseudohyphal transition in *S. cerevisiae* in a cAMP-PKA independent manner and the yeast to hyphal transition in *C. albicans*, both of which have been shown to be crucial for their nutrient foraging ability in diverse niches and the ability of *C. albicans* to successfully infect a host (Palková and Váchová, 2006; Thompson et al., 2011). Comparative transcriptomic analysis identified a conserved metabolic network crucial for these morphogenetic switching events under nitrogen-limiting conditions. Importantly, we observed that perturbations in glycolytic flux using pharmacological and genetic means, negatively impact the expression of genes involved in sulfur metabolism. Furthermore, our functional studies demonstrated that supplementation with sulfur-containing amino acids (cysteine or methionine), effectively rescued the morphogenetic defects resulting from glycolytic impairment, establishing a novel metabolic axis linking glycolysis and sulfur metabolism in the regulation of fungal morphogenesis. Finally, we investigated the implications of this glycolysis-dependent sulfur metabolism on the virulence of *C. albicans*. Our data demonstrates that a *C. albicans* mutant lacking genes that encode for a crucial glycolytic enzyme, phosphofructokinase-1 (Pfk1), exhibits reduced survival within macrophages and severely attenuated virulence in an *in vivo murine* model of systemic candidiasis which is rescued in response to sulfur supplementation to the host.

Collectively, our research, for the first time, identifies a previously uncharacterized metabolic axis linking glycolysis and sulfur metabolism under conditions that induce fungal morphogenesis and underscores the importance of this metabolic network in regulating fungal morphogenesis across two key fungal species and the ability of the human fungal pathogen, *C. albicans* to establish infection in a host. These findings address a fundamental knowledge gap in the field and contributes significantly to our understanding of a broadly conserved phenomenon that exists across a plethora of fungal species.

Results

Glycolysis is Critical for Pseudohyphal Differentiation in a cAMP-PKA Independent Manner in *S. cerevisiae*

Nitrogen limitation is essential for yeast to pseudohyphal transition in *S. cerevisiae* (Gimeno et al., 1992; Pan et al., 2000). However, the influence of carbon sources on this phenomenon is not completely understood. It has been observed that fermetable carbon sources (sucrose, maltose, glucose, etc.) are important for inducing pseudohyphal differentiation under nitrogen-limiting conditions (Van De Velde and Thevelein, 2008). Conversely, when these were replaced with non-fermentable carbon sources, there was a significant reduction in pseudohyphal differentiation (Strudwick et al., 2010). This indicates that the presence of fermentable carbon sources is critical for pseudohyphal differentiation in *S. cerevisiae*. Crabtree positive organisms like *S. cerevisiae* metabolize glucose primarily *via* glycolysis to meet their growth and energy demands (Barford and Hall, 1979; De Deken, 1966). In order to determine whether the ability of *S. cerevisiae* to metabolize glucose *via* glycolysis is critical for pseudohyphal differentiation under nitrogen-limiting conditions, we spotted *S. cerevisiae* wild-type ∑1278b on nitrogen-limiting media containing 2% glucose (SLAD) with and without sub-inhibitory concentrations of 2-Deoxy-D-Glucose (2DG) and sodium citrate (NaCi), which are well characterized glycolysis inhibitors (Cramer and Woodward, 1952; Evans and Ratledge, 1985; Yoshino and Murakami, 1982) and monitored pseudohyphal differentiation (Fig. 1A). Our data demonstrates that sub-inhibitory concentrations of 2DG and NaCi significantly attenuated the ability of *S. cerevisiae* to undergo pseudohyphal differentiation (Fig. 1B). We then isolated cells from these colonies and quantified the percentage of pseudohyphal cells (cells with a length/width ratio of 2 or more (Schröder et al., 2000)) in these colonies. Addition of 2DG or NaCi resulted in a significant reduction in pseudohyphal cells (Fig. 1C) without compromising the overall growth of *S. cerevisiae* wild-type ∑1278b (Fig. 1G). Another diploid *S. cerevisiae* strain that is known to undergo pseudohyphal differentiation under nitrogen-limiting conditions is the prototrophic CEN.PK strain (Laxman and Tu, 2011). In order to determine if our observations in ∑1278b can be recapitulated in CEN.PK, *S. cerevisiae* wild-type CEN.PK was spotted on SLAD with and without sub-inhibitory concentrations of 2DG or NaCi. Our data clearly demonstrates that 2DG and NaCi strongly inhibit pseudohyphal differentiation (Fig. 1B and C) without compromising the overall growth of *S. cerevisiae* wild-type CEN.PK as well (Fig. 1G).

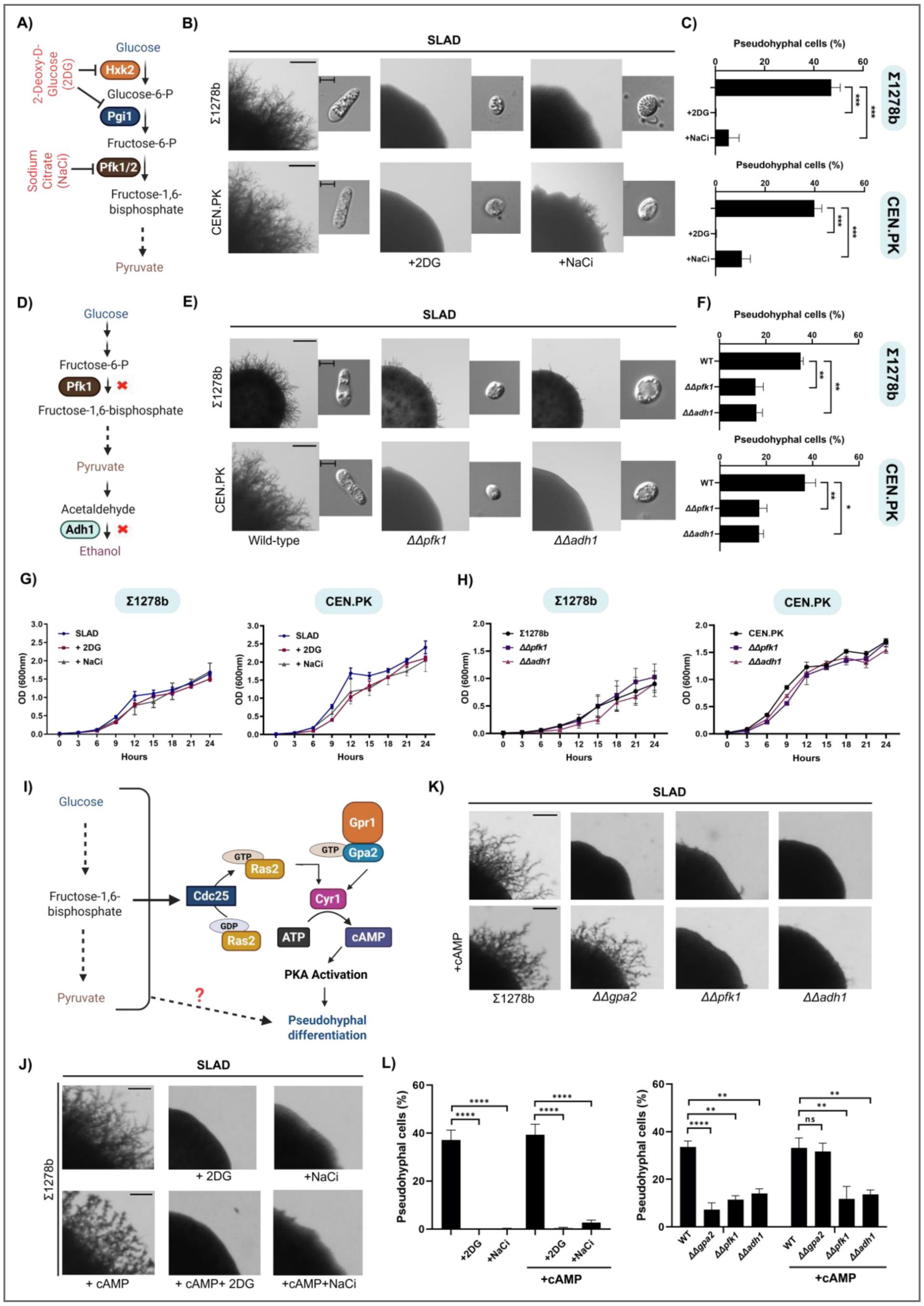


Figure 1. Glycolysis is Critical for Pseudohyphal Differentiation in a cAMP-PKA Independent Manner in *S. cerevisiae*.

(A) Schematic overview of glycolysis and its inhibition by glycolysis inhibitors (2-Deoxy-D-Glucose (2DG) and sodium citrate (NaCi)). **(B)** Wild-type ∑1278b or CEN.PK were spotted on nitrogen-limiting medium containing 2% (w/v) glucose (SLAD) with and without sub-inhibitory concentrations of glycolysis inhibitors (2DG or NaCi), incubated for 10 days at 30 °C. Scale bar represents 1mm for whole colony images and 5μm for single cell images. **(C)** Cells from colonies were isolated and length/width ratio of individual cells was measured using ImageJ and percentage of pseudohyphal cells from total population was determined. More than 500 cells were counted for each condition. Statistical analysis was done using one-way ANOVA test, ***(P<0.001). Error bars represent SEM. **(D)** Schematic overview of glycolysis showing targeted gene deletions. **(E)** Pertinent knockout strains which lack genes encoding for enzymes involved in glycolysis such as *pfk1* and *adh1* were spotted on SLAD along with wild-type ∑1278b or CEN.PK. Scale bar represents 1mm for whole colony images and 5μm for single cell images. **(F)** Cells from colonies were isolated and single cells were imaged using Zeiss Apotome microscope length/width ratio of individual cells was measured using ImageJ and percentage of pseudohyphal cells from total population was determined. More than 500 cells were counted for each strains. Statistical analysis was done using one-way ANOVA test, **(P<0.01) and *(P<0.05). Error bars represent SEM. **(G)** Growth curve was performed to monitor overall growth of wild-type strains on SLAD, SLAD+2DG and SLAD+NaCi. Overnight grown culture of wild-type strains (∑1278b or CEN.PK) were diluted to OD600=0.01 in fresh SLAD medium with and without 2DG (0.05% w/v) or NaCi (0.5% w/v) and allowed to grow at 30 °C for 24 hours. OD600 was recorded at 3-h intervals **(H)** Growth curve was performed to monitor overall growth of wild-type and knockout strains on SLAD. Overnight grown culture of deletion strains (*ΔΔpfk1* or *ΔΔadh1*) along with wild-type strains (∑1278b or CEN.PK) were diluted to OD600=0.01 in fresh SLAD and allowed to grow at 30 °C for 24 hours. OD600 was recorded at 3-h intervals. **(I)** Schematic overview of the role of glucose and glycolytic intermediates in the activation of cAMP-PKA pathway during pseudohyphal differentiation. **(J)** Wild-type ∑1278b was spotted on SLAD containing sub-inhibitory concentration of 2DG or NaCi with and without cAMP (1mM), incubated for 10 days at 30 °C. Scale bar represents 1mm for whole colony images. **(K)** Pertinent knockout strains which lack genes that encode for enzymes involved in glycolysis such as *pfk1* and *adh1* were spotted on SLAD with and without cAMP (1mM), incubated for 10 days at 30 °C. *ΔΔgpa2* strain was used as a control. Scale bar represents 1mm for whole colony images. **(L)** Cells from colonies were isolated and length/width ratio of individual cells was measured using ImageJ and percentage of pseudohyphal cells from total population was determined. More than 500 cells were counted for each condition. Statistical analysis was done using one-way ANOVA test, ****(P<0.0001), **(P<0.01) and ns(non-significant). Error bars represent SEM. This figure was created using *Biorender.com*

Given that, 2DG and NaCi exhibited a strong inhibitory effect on pseudohyphal differentiation under nitrogen-limiting conditions in a strain-independent manner, we wanted to corroborate these observations by using pertinent genetic knockout strains that are compromised in performing glycolysis, in the presence of glucose (Heinisch, 1986; Smith et al., 2004). In order to do this, we generated knockout strains which lack the genes that encode for enzymes that are involved in the glycolysis pathway including *pfk1* (phosphofructokinase-1) and *adh1* (alcohol dehydrogenase-1) (Bennetzen and Hall, 1982; Foy and Bhattacharjee, 1978) (Fig. 1D). Heinisch observed that a null mutant of *pfk1*, a key glycolytic enzyme, is viable and can grow on glucose as a sole carbon source under conditions where cell performs aerobic glycolysis. However, this same mutant exhibits a significant growth defect when cultured anaerobically (Heinisch, 1986; Lobo and Maitra, 1983). Similarly, Dickinson et al., showed that a null mutant of *adh1* is viable. This is likely due to the presence of multiple alcohol dehydrogenase isozymes present in *S. cerevisiae* that likely compensate for the absence of *adh1* (Dickinson et al., 2003). The deletions of *pfk1* and *adh1* genes were confirmed through whole genome sequencing (Suppl.Fig. 1A and B). Our result demonstrates that the deletion of *pfk1* and *adh1* in the ∑1278b background (*ΔΔpfk1* refers to the *pfk1* knockout strain and *ΔΔadh1* refers to the *adh1* knockout strain) resulted in a significant reduction in pseudohyphal differentiation, compared to the corresponding wild-type strain, under nitrogen-limiting conditions (Fig. 1E). We then isolated cells from these colonies and quantified the percentage of pseudohyphal cells in these colonies. Deletion of *pfk1* and *adh1* resulted in a significant reduction in pseudohyphal cells (Fig. 1F) corroborating our data obtained using the glycolytic inhibitors, 2DG and NaCi. Similarly, deletion of *pfk1* and *adh1* in the CEN.PK background resulted in a significant reduction in pseudohyphal differentiation (Fig. 1E and F). In order to rule out whether the reduction in pseudohyphal differentiation in these mutants, is due to a general growth defect, we performed growth curve analysis wherein overnight grown cultures of the deletion strains (*ΔΔpfk1* and *ΔΔadh1)* along with wild-type strains (∑1278b or CEN.PK) were diluted to OD_{600}=0.01 in fresh SLAD and allowed to grow at 30 °C for 24 hours. OD_{600} was recorded at 3-h intervals (Fig. 1H). Our data suggests that the reduction in pseudohyphal differentiation is not due to general growth defect as the mutants were able to grow similar to the wild-type strains in SLAD (Fig. 1H). Taken together, our results clearly demonstrate that the ability of *S. cerevisiae* to efficiently metabolize glucose *via* glycolysis is critical for pseudohyphal differentiation under nitrogen-limiting conditions.

It is well-established that in *S. cerevisiae*, glucose acts as a key signalling molecule and is sensed by the G-protein coupled receptor, Gpr1 that activates the cAMP-PKA pathway, which in turn is crucial for pseudohyphal differentiation under nitrogen-limiting conditions (Lorenz, 1997;

Lorenz et al., 2000). Additionally, a key glycolytic intermediate, fructose-1,6-bisphosphate (FBP), is also known to activate the cAMP-PKA pathway *via* activation of Ras proteins, further linking glucose metabolism to this crucial signalling cascade (Peeters et al., 2017) (Fig. 1I). To investigate whether the pseudohyphal differentiation defects that we observed due to glycolysis perturbation were solely due to the inability of glucose to activate the cAMP-PKA pathway, we performed cAMP add-back assays wherein *S. cerevisiae* wild-type ∑1278b was spotted on SLAD containing sub-inhibitory concentrations of 2DG or NaCi in the presence and absence of 1mM cAMP (Lorenz, 1997). Interestingly, the exogenous addition of cAMP failed to rescue pseudohyphal differentiation defect caused by the perturbation of glycolysis through 2DG or NaCi under nitrogen-limiting condition (Fig. 1J and L). We next asked if the exogenous addition of cAMP rescues the pseudohyphal differentiation defect exhibited by glycolysis defective strains, *ΔΔpfk1* and *ΔΔadh1*. In order to do this, we performed similar cAMP add-back assays wherein *S. cerevisiae* wild-type ∑1278b along with *ΔΔpfk1* and *ΔΔadh1* strains were spotted on SLAD in the presence and absence of 1mM cAMP. The *ΔΔgpa2* strain was used as a control because it is well-established that the deletion of *gpa2* (G protein that activates cAMP production in response to glucose (Rolland et al., 2000)) impairs pseudohyphal differentiation and this defect can be fully rescued by the exogenous addition of cAMP under nitrogen-limiting conditions (Harashima and Heitman, 2002; Lorenz, 1997). Corroborating our previous data, the exogenous addition of cAMP failed to rescue pseudohyphal differentiation defect caused by the perturbation of glycolysis *via* the deletion of *pfk1* and *adh1* but fully rescued the pseudohyphal differentiation defect exhibited by the *ΔΔgpa2* strain (Fig. 1K and L). Taken together, our results clearly demonstrate that the ability of *S. cerevisiae* to efficiently metabolize glucose *via* glycolysis is critical for pseudohyphal differentiation in a cAMP-PKA pathway independent manner under nitrogen-limiting conditions. This implies that glycolysis may be parallelly regulating other cellular processes essential for this morphological transition under nitrogen-limiting conditions.

Comparative Transcriptomics Identifies Glycolysis-dependent Regulation of Sulfur Metabolism During Pseudohyphal Differentiation in *S. cerevisiae*

Our previous data showed that the perturbation of glycolysis using specific inhibitors including 2DG/NaCi or deletion of genes that encode for glycolytic enzymes including *pfk1/ adh1* resulted in a significant reduction in pseudohyphal differentiation in a cAMP-PKA independent manner, under nitrogen-limiting conditions (Fig. 1). Given that, glycolysis strongly influences the global transcriptional landscape of *S. cerevisiae* (Newcomb et al., 2003; Wu et al., 2019), we undertook a comprehensive transcriptomic approach to identify the changes in the transcriptional landscape of *S. cerevisiae* colonies, in conditions that perturb glycolysis, resulting in the inhibition of pseudohyphal differentiation. We performed comparative RNA-Seq analysis using RNA isolated from wild-type cells spotted on SLAD in the presence and absence of the glycolysis inhibitor, 2DG (2DG addition was used to perturb glycolysis in our transcriptomics experiments, as it exerted the strongest inhibitory effect on pseudohyphal differentiation as shown in Fig. 1B and C). The experimental schematic is depicted in Fig. 2A. Briefly, comparative RNA-Seq analysis was done using wild-type colonies grown on SLAD in the presence and absence of 2DG isolated on day 5 (D-5) and day 10 (D-10), respectively (Fig. 2A). We have used D-5 as an early time point for our experiments, as we have consistently observed the emergence of pseudohyphae in colonies spotted on SLAD at this time-point and D-10 as a late time point wherein colonies exhibit robust pseudohyphal differentiation (Suppl.Fig. 2A). A volcano plot of differential gene expression in 2DG treated cells compared to untreated cells in both D-5 and D-10 is shown in Fig. 2B wherein some of the significantly upregulated and downregulated genes are highlighted with green colour. Previous studies in the field have shown that, under nitrogen-limiting/glucose-replete conditions, glycolytic intermediates provide various precursors needed for the biosynthesis of several amino acids in order to maintain amino acid homeostasis (Dikicioglu et al., 2011; Martíez-force and Benítez, 1992) and perturbation of glycolysis under these conditions results in the increased transcription of various genes involved in amino acid biosynthesis, as a feedback response. As expected, the expression of several genes involved in amino acid biosynthesis and transport were significantly upregulated in 2DG treated cells compared to untreated cells, in both D-5 and D-10 respectively (Suppl.Fig. 2B) corroborating the observations from the previously mentioned studies (Dikicioglu et al., 2011; Martíez-force and Benítez, 1992; Wu et al., 2004). Interestingly and rather unexpectedly, the expression of multiple genes specifically involved in the biosynthesis and transport of sulfur-containing amino acids, cysteine and methionine were significantly downregulated in 2DG treated cells compared to untreated cells in both D-5 and D-10 respectively (Fig. 2C). Our RNA-Seq analysis clearly demonstrates that multiple genes involved in the de novo biosynthesis of sulfur-containing amino acids, are significantly downregulated in the presence of 2DG, under nitrogen-limiting conditions.

Given that, perturbation of glycolysis *via* 2DG addition resulted in a clear downregulation of multiple genes involved in the *de novo* biosynthesis of sulfur-containing amino acids under nitrogen-limiting conditions, we wanted to corroborate these observations by using pertinent genetic knockout strains, *ΔΔpfk1* and *ΔΔadh1*. In order to determine this, we spotted *S. cerevisiae* wild-type ∑1278b along with *ΔΔpfk1* and *ΔΔadh1* on SLAD and cells from colonies were isolated after ~4 days for RNA isolation. We then performed comparative RT-qPCR to check for the relative expression of genes involved in *de novo* biosynthesis of sulfur-containing amino acids including *met32* (Zinc-finger DNA-binding transcription factor), *met3* (ATP sulfurylase), *met5* (Sulfite reductase beta subunit), and *met17* (O-acetyl homoserine-O-acetyl serine sulfhydrylase) (Blaiseau et al., 1997; Masselot and de Robichon-Szulmajster, 1975). Our data clearly shows that all the aforesaid genes were significantly downregulated in the *ΔΔpfk1* and *ΔΔadh1* strains compared to wild-type (Suppl.Fig. 2C) suggesting that genetic perturbation of glycolysis negatively impacts sulfur metabolism under nitrogen-limiting conditions.

In order to determine whether these observed trends are reflected at the protein level, we generated strains wherein various proteins involved in the de novo biosynthesis of sulfur-containing amino acids including Met4 (Leucine-zipper transcriptional activator), Met32 (Zinc-finger DNA-binding transcription factor), Met16 (3'-phosphoadenylsulfate reductase), Met10 (Subunit alpha of assimilatory sulfite reductase), Cys4 (Cystathionine beta-synthase) and Cys3 (Cystathionine gamma-lyase) were epitope-tagged (Fig. 2D). These strains were spotted on SLAD in the presence and absence of 2DG following which, cells from colonies were isolated on D-5 and levels of these proteins were assessed by using Western blotting (Fig. 2E). Our results indicate that the levels of Met4, Met32, Met16, Met10, Cys4 and Cys3 were significantly reduced in the 2DG treated cells compared to untreated cells, corroborating our transcriptome data (Fig. 2E and Suppl.Fig. 3A). This clearly demonstrates that perturbation of glycolysis negatively influences the expression of sulfur metabolism proteins, under nitrogen-limiting conditions. We next asked, if the glycolysis-dependent regulation of sulfur metabolism under nitrogen-limiting conditions (SLAD) is restricted to conditions wherein cells physically attach to solid surfaces (colony growth) or whether it is conserved even in conditions where cells are free-floating (liquid-culture growth). In order to do this, pertinent epitope-tagged strains (wherein Met32, Met16, Met10 and Cys3 were HA-tagged) were grown in liquid SLAD in the presence and absence of 2DG for 24 hours after which the expression of the aforesaid proteins were assessed using Western blotting (Suppl.Fig. 3B and C). Importantly, our results indicate that the levels of Met32, Met16, Met10 and Cys3 were significantly reduced in the 2DG treated cells compared to untreated cells (Suppl.Fig. 3B and C). Taken together, our data clearly demonstrates that the perturbation of glycolysis under nitrogen-limiting conditions negatively influences sulfur metabolism.

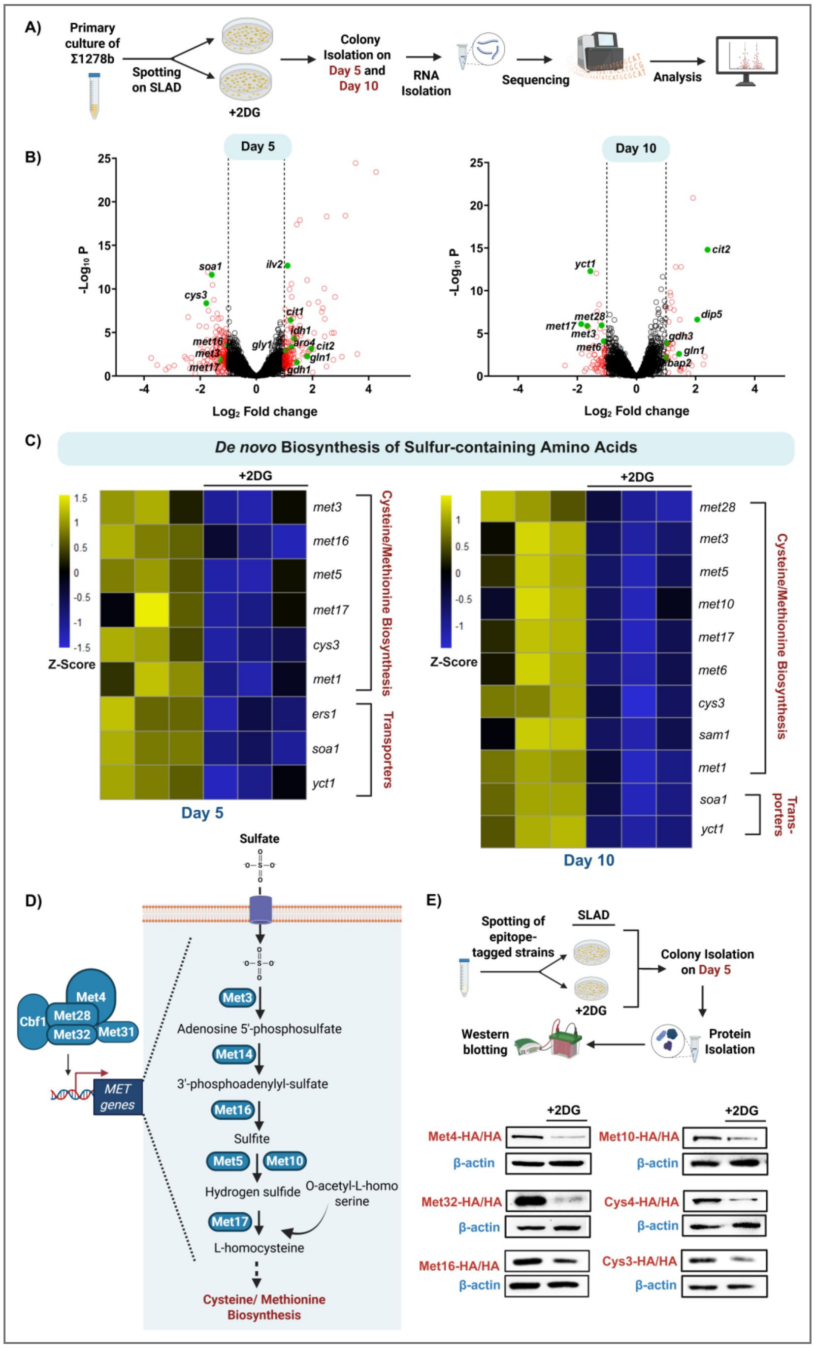


Figure 2. Comparative Transcriptomics Identifies the Role of Glycolysis in Regulating Sulfur Metabolism During Pseudohyphal Differentiation in *S. cerevisiae*.

(A) Schematic overview of steps involved in RNA isolation and sequencing. **(B)** Volcano plots represent differentially expressed genes in 2DG treated cells compared to untreated cells isolated from D-5 and D-10 colonies, respectively. Genes which are upregulated (Log2 fold change ≥ 1) and downregulated (Log2 fold change ≤ −1) are highlighted in red. Some of the significantly upregulated and downregulated genes involved in the biosynthesis of various amino acids are highlighted in green. **(C)** Heatmaps represent the differentially expressed genes involved in *de novo* biosynthesis and transport of sulfur-containing amino acids, in 2DG treated cells compared to untreated cells isolated from D-5 and D-10 colonies, respectively (n=3). Scale bar represents Z-Score. **(D)** Schematic overview of *de novo* biosynthesis of sulfur-containing amino acids. **(E)** Schematic overview of steps involved in the Western blotting experiments. Epitope-tagged strains of various proteins involved in the *de novo* biosynthesis of sulfur-containing amino acids including Met4, Met32, Met16, Met10, Cys4 and Cys3 were spotted on SLAD in the presence and absence of sub-inhibitory concentration of 2DG following which, cells from colonies were isolated on D-5 and levels of these proteins were assessed using Western blotting. β-actin was used as loading control. This figure was created using *Biorender.com*

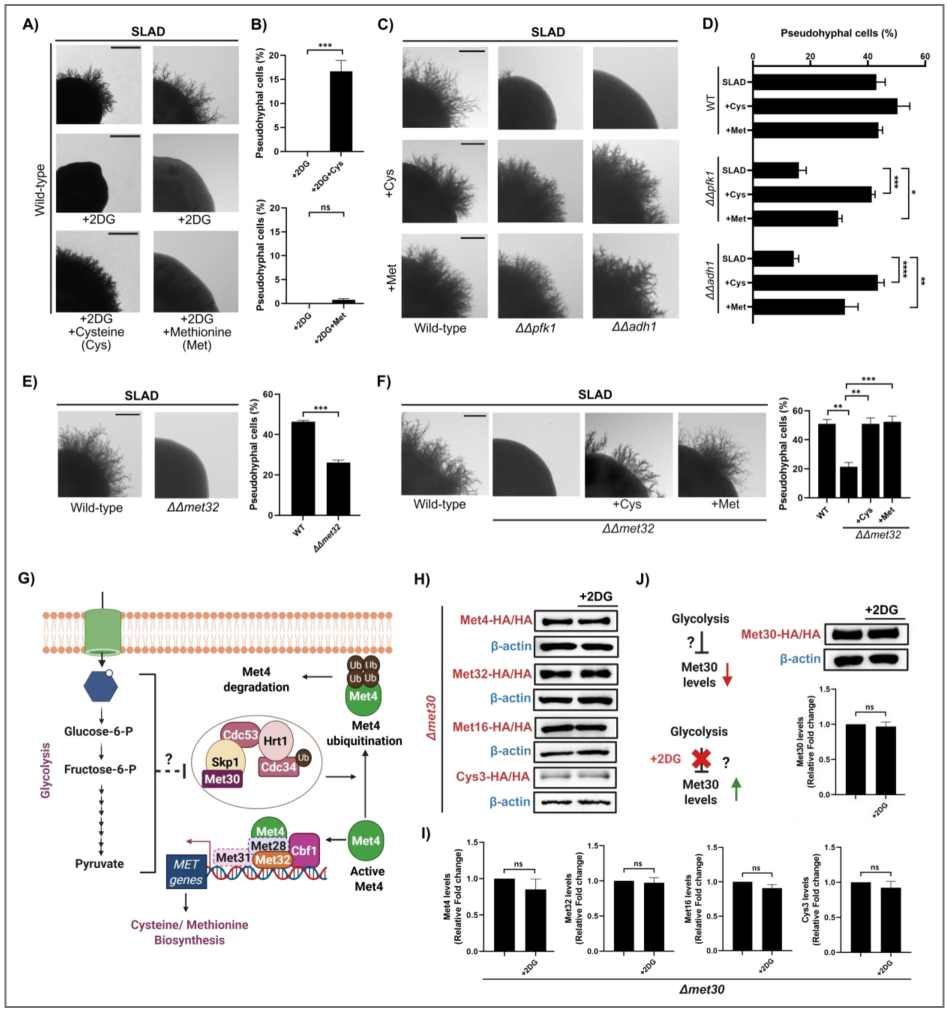


Figure 3. **Glycolysis-mediated Regulation of Sulfur Metabolism is Critical for Pseudohyp-hal Differentiation in *S. cerevisiae*.**

(A) Wild-type ∑1278b was spotted on SLAD containing sub-inhibitory concentration of 2DG with and without sulfur-containing compounds including cysteine (500µM) or methionine (20µM), incubated for 10 days at 30 °C. Scale bar represents 1mm for whole colony images. **(B)** Cells from colonies were isolated and length/width ratio of individual cells was measured using ImageJ and percentage of pseudohyphal cells from total population was determined. More than 500 cells were counted for each condition. Statistical analysis was done using unpaired t-test, ***(P<0.001) and ns (non-significant). Error bars represent SEM. **(C)** Pertinent knockout strains (*ΔΔpfk1* or *ΔΔadh1*) along with wild-type were spotted on SLAD with and without sulfur-containing compounds including cysteine (200µM) or methionine (20µM), incubated for 10 days at 30 °C. Scale bar represents 1mm for whole colony images. **(D)** Cells from colonies were isolated and length/width ratio of individual cells was measured using ImageJ and percentage of pseudohyphal cells from total population was determined. More than 500 cells were counted for each condition. Statistical analysis was done using one-way ANOVA test, ****(P<0.0001), ***(P<0.001), **(P<0.01) and *(P<0.05). Error bars represent SEM. **(E)** Pertinent knockout strains which lack genes encoding for transcription factor involved in *de novo* biosynthesis of sulfur-containing amino acids including *met32* along with wild-type ∑1278b were spotted on SLAD. Scale bar represents 1mm for whole colony images. Cells from colonies were isolated and length/width ratio of individual cells was measured using ImageJ and percentage of pseudohyphal cells from total population was determined. More than 500 cells were counted for each condition. Statistical analysis was done using unpaired t-test, ***(P<0.001). Error bars represent SEM. **(F)** Pertinent knockout strain (*ΔΔmet32)* was spotted on SLAD with and without sulfur-containing compounds including cysteine (200µM) or methionine (20µM), incubated for 10 days at 30 °C. Wild-type spotted on SLAD was used as control. Scale bar represents 1mm for whole colony images. Cells from colonies were isolated and length/width ratio of individual cells was measured using ImageJ and percentage of pseudohyphal cells from total population was determined. More than 500 cells were counted for each condition. Statistical analysis was done using one-way ANOVA test, ***(P<0.001) and **(P<0.01). Error bars represent SEM. **(G)** Schematic overview of the proposed role of SCF^{Met30} in glycolysis-dependent regulation of sulfur metabolism during pseudohyphal differentiation under nitrogen-limiting conditions in *S. cerevisiae.***(H)** Epitope-tagged strains of various proteins involved in the *de novo* biosynthesis of sulfur-containing amino acids including Met4, Met32, Met16 and Cys3 in *Δmet30* background were spotted on SLAD in the presence and absence of sub-inhibitory concentration of 2DG following which, cells from colonies were isolated on D-5 and levels of these proteins were assessed using Western blotting. β-actin was used as loading control. **(I)** Raw images of Western blots were analysed using ImageJ to normalize targeted protein expression with the expression of housekeeping protein-β-actin, in order to generate densitometric graphs. Statistical analysis was done using unpaired t-test, ns (non-significant). Error bars represent SEM. **(J)** Epitope-tagged strain of Met30 was spotted on SLAD in the presence and absence of sub-inhibitory concentration of 2DG following which, cells from colonies were isolated on D-5 and levels of these proteins were assessed using Western blotting. β-actin was used as loading control. Raw images of Western blots were analysed using ImageJ to normalize targeted protein expression with the expression of housekeeping protein-β-actin, in order to generate densitometric graphs. Statistical analysis was done using unpaired t-test, ns (non-significant). Error bars represent SEM. This figure was created using *Biorender.com*

Exogenous Supplementation of Sulfur Sources Rescues Pseudohyphal Differentiation Defects Caused by the Perturbation of Glycolysis in *S. cerevisiae*

Our comparative transcriptomics data clearly demonstrates that, under nitrogen-limiting conditions, glycolysis plays a critical role in regulating the expression of genes involved in the *de novo* biosynthesis of sulfur-containing amino acids (Fig. 2). Based on our previous data, we hypothesized that under nitrogen-limiting conditions, active glycolysis enables *de novo* biosynthesis of sulfur-containing amino acids which in turn, is critical for pseudohyphal differentiation and perturbation of active glycolysis using inhibitors like 2DG or disruption of genes involved in glycolysis including *pfk1* or *adh1* negatively affects glycolysis-mediated regulation of sulfur metabolism leading to attenuation of pseudohyphal differentiation. If the proposed hypothesis holds true, the exogenous supplementation of sulfur-containing compounds should mitigate the fungal morphogenetic defects resulting from the perturbation of glycolysis. In order to test this, we performed sulfur add-back assays wherein *S. cerevisiae* wild-type ∑1278b was spotted on SLAD containing sub-inhibitory concentration of 2DG in the presence and absence of sulfur-containing amino acids (cysteine or methionine). Remarkably, the exogenous addition of cysteine resulted in a significant rescue of the pseudohyphal differentiation defect observed in nitrogen-limiting medium containing 2DG (Fig. 3A). Corroborating this observation, the percentage of pseudohyphal cells in colonies isolated from SLAD containing 2DG and cysteine were significantly higher than the percentage of pseudohyphal cells in colonies isolated from SLAD containing 2DG alone (Fig. 3B). Interestingly, the addition of methionine failed to rescue the pseudohyphal differentiation defects caused by the perturbation of glycolysis (Fig. 3A and B).

We next asked if the exogenous addition of sulfur sources rescues the pseudohyphal differentiation defect exhibited by glycolysis defective strains, *ΔΔpfk1* and *ΔΔadh1*. In order to do this, we performed similar sulfur add-back assays wherein *S. cerevisiae* wild-type ∑1278b along with *ΔΔpfk1* and *ΔΔadh1* strains were spotted on SLAD in the presence and absence of sulfur-containing amino acids (cysteine or methionine). The exogenous addition of cysteine and methionine were able to significantly rescue the pseudohyphal differentiation defect exhibited by the *ΔΔpfk1* and *ΔΔadh1* strains albeit at different levels (Fig. 3C). Corroborating this observation, the percentage of pseudohyphal cells in *ΔΔpfk1* and *ΔΔadh1* colonies isolated from SLAD containing cysteine or methionine were significantly higher than the percentage of pseudohyphal cells in *ΔΔpfk1* and *ΔΔadh1* colonies isolated from SLAD alone (Fig. 3D). Taken together, our results clearly demonstrate that the glycolysis-mediated regulation of sulfur metabolism is critical for pseudohyphal differentiation under nitrogen-limiting conditions.

Sulfur Metabolism is Critical for Pseudohyphal Differentiation Under Nitrogen-limiting Conditions in *S. cerevisiae*

Our previous observations strongly suggest a direct role for sulfur metabolism in regulating fungal morphogenesis, under nitrogen-limiting conditions. In order to determine whether glycolysis-dependent sulfur metabolism is critical for pseudohyphal differentiation, under nitrogen-limiting conditions, we generated a knockout strain that lacks *met32* (gene that encodes for a zinc-finger DNA-binding transcription factor involved in the regulation of *de novo* biosynthesis of sulfur-containing amino acids (Blaiseau et al., 1997)). Unlike deletions of most other genes involved in the *de novo* biosynthesis of sulfur-containing amino acids, *met32* deletion strain does not exhibit auxotrophy for methionine or cysteine (Blaiseau et al., 1997). This is likely due to the functional redundancy provided by its paralog, *met31* (Blaiseau et al., 1997). We then assessed the ability of the *met32* deletion strain to undergo pseudohyphal differentiation under nitrogen-limiting conditions. Deletion of *met32* resulted in a significant reduction in pseudohyphal differentiation. Corroborating this observation, the percentage of pseudohyphal cells in *ΔΔmet32* colonies were significantly lower in SLAD, compared to *S. cerevisiae* wild-type ∑1278b colonies (Fig. 3E). This suggests that perturbations to sulfur metabolism negatively affects pseudohyphal differentiation. In order to rule out whether the reduction in pseudohyphal differentiation in this mutant, is due to a general growth defect, we performed growth curve analysis wherein overnight grown cultures of deletion strain (*ΔΔmet32)* along with wild-type strain (∑1278b) were diluted to OD_{600}=0.01 in fresh SLAD and allowed to grow at 30 °C for 24 hours. OD_{600} was recorded at 3-h intervals (Suppl.Fig. 4A). Our data suggests that the reduction in pseudohyphal differentiation is not due to general growth defect as the mutant was able to grow similar to the wild-type strain in SLAD. We next asked if the exogenous addition of sulfur sources rescues the pseudohyphal differentiation defect exhibited by *ΔΔmet32*. In order to do this, we performed a simple sulfur add-back assay, wherein *ΔΔmet32* strain was spotted on SLAD containing either cysteine or methionine and allowed to undergo pseudohyphal differentiation. Our results show that the addition of cysteine or methionine completely rescued the pseudohyphal differentiation defect exhibited by *ΔΔmet32* strain. The percentage of pseudohyphal cells were also significantly higher in *ΔΔmet32* colonies growing on SLAD containing cysteine or methionine compared to just SLAD (Fig. 3F). Taken together, our data clearly demonstrates that the efficient *de novo* biosynthesis of sulfur-containing amino acids including cysteine and methionine is critical for pseudohyphal differentiation, under nitrogen-limiting conditions.

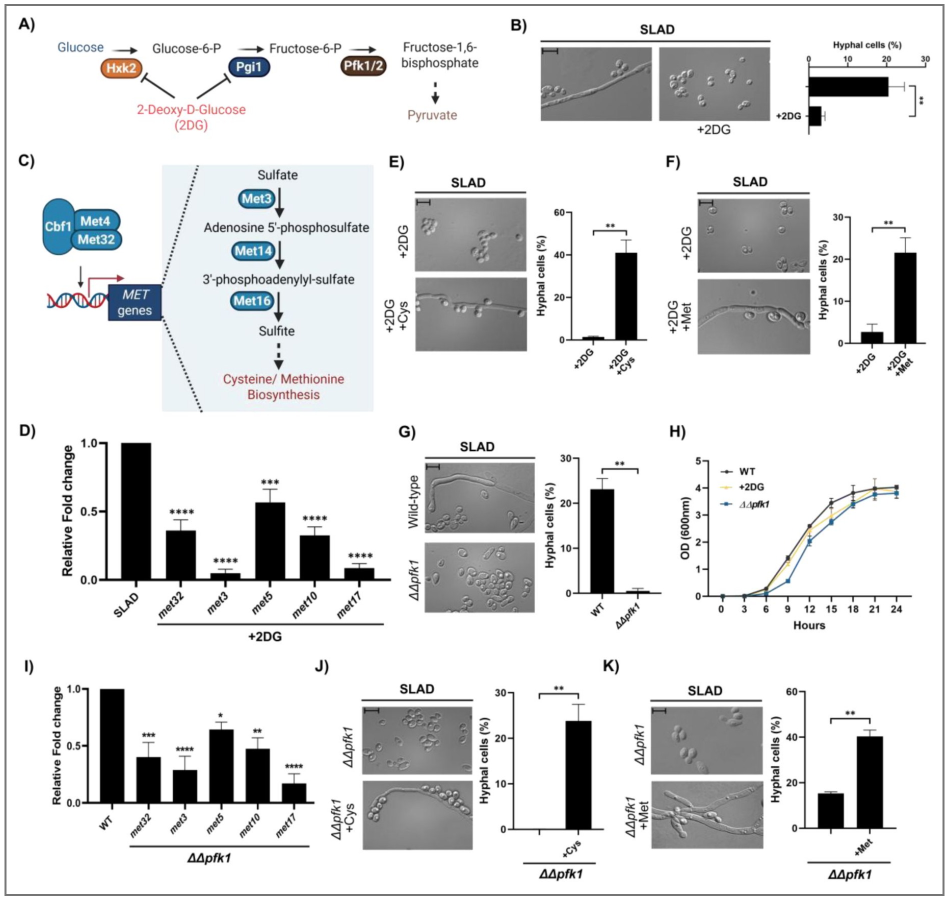


Figure 4. Glycolysis-dependent Sulfur Metabolism is Critical for Hyphal Differentiation in *C. albicans*.

(A) Schematic overview of glycolysis and its inhibition by glycolysis inhibitor (2DG). **(B)** Wild-type SC5314 was spotted on SLAD with and without sub-inhibitory concentration of glycolysis inhibitor 2DG, incubated for 7 days at 37 °C. Cells from colonies were isolated and length/width ratio of individual cells was measured using ImageJ and percentage of hyphal cells from total population was determined. More than 500 cells were counted for each condition. Scale bar represents 10µm for single cell images. Statistical analysis was done using unpaired t-test, **(P<0.01). Error bars represent SEM. **(C)** Schematic overview of *de novo* biosynthesis of sulfur-containing amino acids. **(D)** Wild-type SC5314 was spotted on SLAD, in the presence and absence of sub-inhibitory concentration of 2DG and cells from colonies were isolated after ~4 days, for RNA isolation. We then performed comparative RT-qPCR to check for the relative expression of genes involved in the *de novo* biosynthesis of sulfur-containing amino acids pathway including *met32*, *met3*, *met5* (*ecm17*), *met10* and *met17* (*met15*). Statistical analysis was done using one-way ANOVA test, ****(P<0.0001) and ***(P<0.001). Error bars represent SEM. **(E)** Wild-type SC5314 was spotted on SLAD containing sub-inhibitory concentration of 2DG with and without cysteine (100µM) and, incubated for 7 days at 37 °C. Cells from colonies were isolated and length/width ratio of individual cells was measured using ImageJ and percentage of hyphal cells from total population was determined. Scale bar represents 10µm for single cell images. More than 500 cells were counted for each condition. Statistical analysis was done using unpaired t-test, **(P<0.01). Error bars represent SEM. **(F)** Wild-type SC5314 was spotted on SLAD containing sub-inhibitory concentration of 2DG with and without methionine (20µM) and, incubated for 7 days at 37 °C. Cells from colonies were isolated and length/width ratio of individual cells was measured using ImageJ and percentage of hyphal cells from total population was determined. More than 500 cells were counted for each condition. Scale bar represents 10µm for single cell images. Statistical analysis was done using unpaired t-test, **(P<0.01). Error bars represent SEM. **(G)** Wild-type SC5314 and pertinent knockout strain which lacks the genes encoding for a glycolytic enzyme, *pfk1* (*ΔΔpfk1*) were spotted on SLAD, incubated for 7 days at 37 °C. Cells from colonies were isolated and length/width ratio of individual cells was measured using ImageJ and percentage of hyphal cells from total population was determined. Scale bar represents 10µm for single cell images. More than 500 cells were counted for each strain. Statistical analysis was done using unpaired t-test, **(P<0.01). Error bars represent SEM. **(H)** Growth curve was performed to monitor overall growth of wild-type strain on SLAD and SLAD+2DG and *ΔΔpfk1* on SLAD. Overnight grown culture of wild-type strain (SC5314) were diluted to OD600=0.01 in fresh SLAD medium with and without 2DG (0.2% (w/v)) and allowed to grow at 30 °C for 24 hours. OD600 was recorded at 3-h intervals. For *ΔΔpfk1*, overnight grown culture of *ΔΔpfk1* strains along with wild-type strain (SC5314) were diluted to OD600=0.01 in fresh SLAD and allowed to grow at 30 °C for 24 hours. OD600 was recorded at 3-h intervals. **(I)***ΔΔpfk1* along with Wild-type SC5314 was spotted on SLAD and cells from colonies were isolated after ~4 days, for RNA isolation. We then performed comparative RT-qPCR to check for the relative expression of genes involved in the *de novo* biosynthesis of sulfur-containing amino acids pathway including *met32*, *met3*, *met5* (*ecm17*), *met10* and *met17* (*met15*). Statistical analysis was done using one-way ANOVA test, **** (P<0.0001), ***(P<0.001), **(P<0.01) and *(P<0.05). Error bars represent SEM. **(J)***ΔΔpfk1* was spotted on SLAD in the presence and absence of cysteine (100μM), incubated for 7 days at 37 °C. Cells from colonies were isolated and length/width ratio of individual cells was measured using ImageJ and percentage of hyphal cells from total population was determined. More than 500 cells were counted for each condition. Statistical analysis was done using unpaired t-test, **(P<0.01). Error bars represent SEM. **(K)***ΔΔpfk1* was spotted on SLAD in the presence and absence of methionine (10μM), incubated for 7 days at 37 °C. Cells from colonies were isolated and length/width ratio of individual cells was measured using ImageJ and percentage of hyphal cells from total population was determined. More than 500 cells were counted for each condition. Statistical analysis was done using unpaired t-test, **(P<0.01). Error bars represent SEM. This figure was created using *Biorender.com*

Met30 is Involved in the Glycolysis-dependent Regulation of Sulfur Metabolism During Pseudohyphal Differentiation in *S. cerevisiae*

Our previous data clearly demonstrates that, under nitrogen-limiting conditions, the levels of Met4, a key activator of genes involved in the *de novo* biosynthesis of sulfur-containing amino acids, was significantly reduced in 2DG treated cells compared to untreated cells (Fig. 2E). Met4 activity is known to be negatively regulated by SCF^{Met30} (Skp1/Cullin1/F-box) ubiquitin ligase complex in *S. cerevisiae* (Rouillon et al., 2000). Under sulfur-replete conditions, Met30 promotes Met4 ubiquitination and degradation, thereby repressing the *MET* gene network. Conversely, sulfur limitation reduces Met30 activity, allowing Met4 to become active and induce the expression of genes required for sulfur amino acid biosynthesis (Smothers et al., 2000). It is important to note that deletion of *met30* is lethal in haploid strains of *S. cerevisiae* (Kaiser et al., 1998). While point mutations or temperature sensitive *met30* mutants are commonly used in haploid strains to circumvent this lethality (Kaiser et al., 1998; Su et al., 2008), our study in diploid *S. cerevisiae* involved generating a heterozygous viable *met30* (*Δmet30*) deletion strain. In order to confirm that the deletion of a single copy of *met30* does not lead to any growth defects in SLAD, we performed growth curve analysis wherein overnight grown cultures of deletion strain (*Δmet30)* along with wild-type strain (∑1278b) were diluted to OD_{600}=0.01 in fresh SLAD and allowed to grow at 30 °C for 24 hours. OD_{600} was recorded at 3-h intervals (Suppl.Fig. 4A). Our data suggests that the deletion of *met30* does not affect overall growth as the mutant was able to grow similar to the wild-type strain in SLAD (Suppl.Fig. 4A). Based on our previous data, we hypothesized that under nitrogen-limiting conditions, active glycolysis enables *de novo* biosynthesis of sulfur-containing amino acids by reducing the activity of Met30. Conversely, we hypothesized that perturbing active glycolysis with glycolysis inhibitor, 2DG increases Met30 activity, which in turn inactivates Met4, thereby negatively affecting *MET* gene transcription, ultimately resulting in attenuated pseudohyphal differentiation (Fig. 3G). In order to investigate this hypothesis, we generated strains wherein one copy of *met30* was deleted and various proteins involved in the *de novo* biosynthesis of sulfur-containing amino acids including Met4 (Leucine-zipper transcriptional activator), Met32 (Zinc-finger DNA-binding transcription factor), Met16 (3'-phosphoadenylsulfate reductase) and Cys3 (Cystathionine gamma-lyase) were epitope-tagged. These strains were spotted on SLAD in the presence and absence of 2DG following which, cells from colonies were isolated on D-5 and levels of these proteins were assessed by using Western blotting. Our results indicate that the levels of Met4, Met32, Met16 and Cys3 were similar in both 2DG treated and untreated cells, in the *Δmet30* deletion strain (Fig. 3H and I). This suggests that glycolysis influences Met30 activity which in turn regulates Met4 activity thereby affecting the expression of genes involved in the *de novo* biosynthesis of sulfur-containing amino acids, under nitrogen-limiting conditions.

Based on our previous observation that 2DG treatment significantly reduces Met4 levels in nitrogen-limiting conditions, (an effect that was notably absent in *met30* deletion strains), we hypothesized that the 2DG-induced downregulation of Met4 is mediated by an increase in Met30 levels. In order to check Met30 levels in the presence of 2DG, we generated a strain wherein Met30 was epitope-tagged and this strain was spotted on SLAD, in the presence and absence of 2DG following which, cells from colonies were isolated on D-5 and levels of Met30 were assessed by using Western blotting. Our results indicate that the levels of Met30 were similar in both 2DG treated and untreated cells (Fig. 3J), suggesting that glycolysis influences Met30 activity post-translationally rather than through changes in its protein abundance. Collectively, our findings indicate that the perturbation of glycolysis under nitrogen-limiting conditions negatively impacts sulfur metabolism through a mechanism dependent on Met30 activity.

Glycolysis-dependent Sulfur Metabolism is Critical for Hyphal Differentiation in C. albicans

C. albicans is an opportunistic fungal pathogen that exhibits a remarkable degree of fungal morphogenesis in response to various environmental cues (Huang, 2012). Yeast to hyphal differentiation in *C. albicans* is well studied. However, we do not have a complete understanding of fundamental metabolic drivers that orchestrate this phenomenon. Given that central carbon metabolic pathways including glycolysis are well conserved across a plethora of fungal species including *C. albicans* (Askew et al., 2009) and nitrogen limitation is one of the key triggers for yeast to hyphal differentiation in *C. albicans* (Csank et al., 1998), we wanted to determine if glycolysis-mediated sulfur metabolism plays an important role in this morphogenetic switching in *C. albicans* as well. First, we wanted to understand whether the ability of C. *albicans* cells to metabolize glucose, under nitrogen-limiting conditions is critical for fungal morphogenesis (Fig. 4A). In order to do this, *C. albicans* wild-type SC5314 was spotted on SLAD, in the presence and absence of sub-inhibitory concentration of glycolysis inhibitor, 2DG, allowed to grow for 7 days, and cells were isolated and imaged using bright-field microscopy. Our data clearly demonstrates that sub-inhibitory concentration of 2DG significantly reduced filamentation in *C. albicans* (Fig. 4B) without compromising the overall growth of *C. albicans* wild-type SC5314 (Fig. 4H). We then isolated cells from these colonies and quantified the percentage of hyphal cells (cells with a length/width ratio of 4.5 or more (Su et al., 2018)) in these colonies. Corroborating our microscopy data, the percentage of hyphal cells in colonies spotted on SLAD containing 2DG was significantly lower compared to the percentage of hyphal cells in colonies spotted on SLAD without 2DG (Fig. 4B). To investigate whether the hyphal differentiation defects that we observed due to glycolysis perturbation were solely due to the inability of glucose to activate the cAMP-PKA pathway, we performed cAMP add-back assays wherein *C. albicans* wild-type SC5314 was spotted on SLAD containing sub-inhibitory concentrations of 2DG in the presence and absence of 5mM cAMP. Interestingly, corroborating our *S. cerevisiae* data, the exogenous addition of cAMP failed to rescue hyphal differentiation defect caused by the perturbation of glycolysis through 2DG under nitrogen-limiting condition in *C. albicans* (Suppl.Fig. 5B). Taken together, our results show that the ability of *C. albicans* to metabolize glucose is critical for fungal morphogenesis in a cAMP-PKA pathway independent manner under nitrogen-limiting conditions.

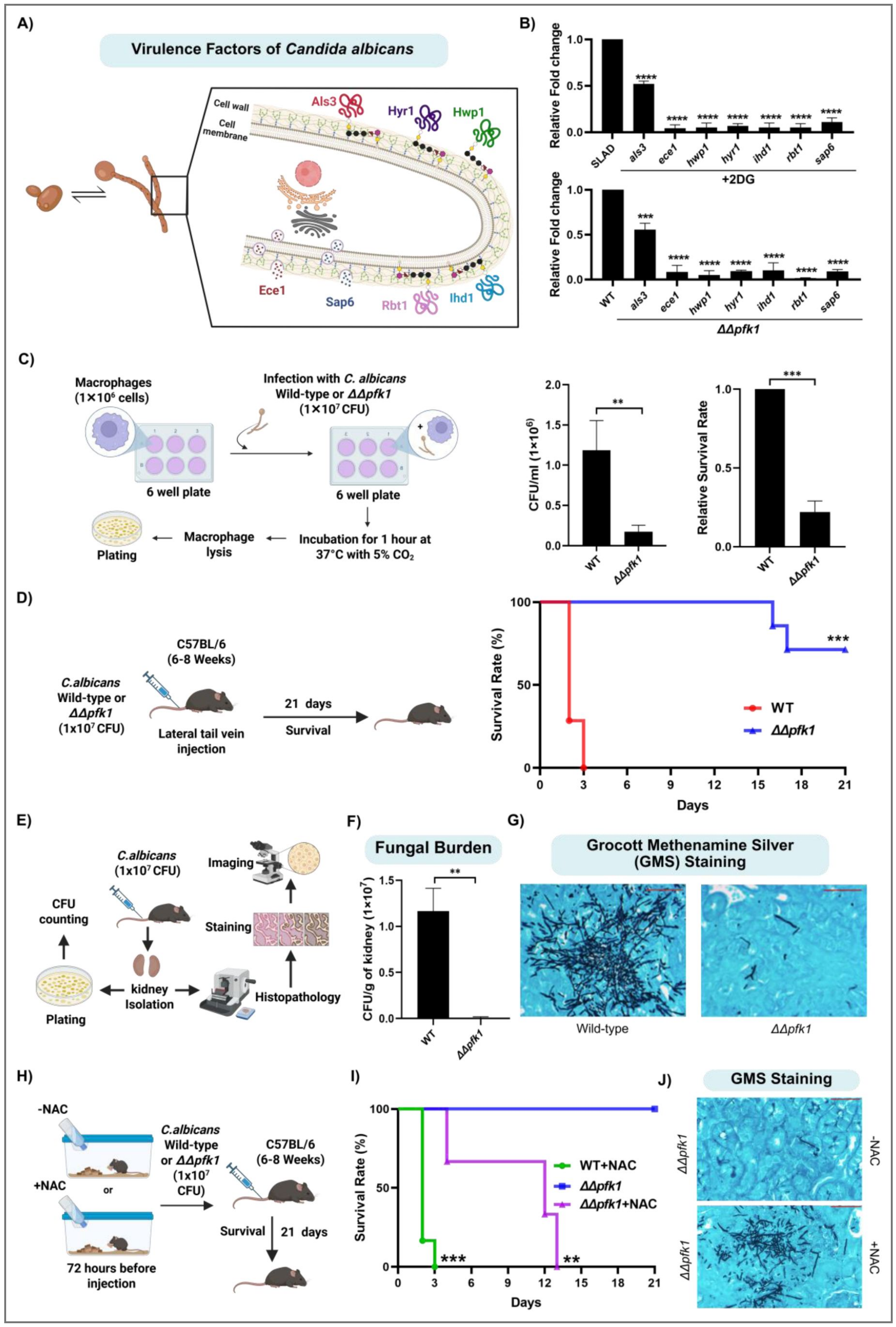


Figure 5. **Perturbation of Glycolysis Leads to Attenuated Fungal Virulence in *C. albicans*.**

(A) Schematic overview of various proteins involved in hyphal differentiation and virulence of *C. albicans*. **(B)** Wild-type SC5314 was spotted on SLAD, in the presence and absence of sub-inhibitory concentration of 2DG or wild-type SC5314 and *ΔΔpfk1* were spotted on SLAD and cells from these colonies were isolated after ~4 days, for RNA isolation. We then performed comparative RT-qPCR to check for the relative expression of genes involved in hyphal differentiation and virulence of *C. albicans* including *als3, ece1, hwp1, hyr1, ihd1, rbt1* and *sap6.* Statistical analysis was done using one-way ANOVA test, ****(P<0.0001) and ***(P<0.001). Error bars represent SEM. **(C)** Schematic overview of *in vitro* microbial survival assay. *RAW 264.7* macrophages were incubated with the wild-type and *ΔΔpfk1* strain with MOI=10. After 1 hour of incubation, macrophages were lysed and plating was done to enumerate CFU. Percentage of survival was expressed as the number of CFU in the presence of macrophages divided by the number of CFU in the absence of macrophages. Statistical analysis was done using unpaired t-test, ***(P<0.001) and **

(P<0.01). Error bars represent SEM. **(D)** 6-8-week-old C57BL/6 mice were infected intravenously *via* lateral tail vein injection with 1 × 10^7 CFU of wild-type SC5314 or 1 × 10^7 CFU of *ΔΔpfk1*. To determine survival rate, mice were monitored for 21 days post-infection for clinical signs of illness or mortality. The results are representative of seven mice per injected strain. Survival percentage were statistically evaluated by the Log-rank Mantel-Cox test, *** (P<0.001) (n=7). **(E)** Schematic overview of kidney isolation to check fungal burden using CFU counting and histology. **(F)** Fungal burden was measured by CFU counting after kidney isolation from mice injected with either wild-type or *ΔΔpfk1*. Statistical analysis was done using unpaired t-test, **(P<0.01). Error bars represent SEM. **(G)** Histopathological analysis of kidney sections using Grocott Methanamine Silver (GMS) staining was done for kidneys isolated from mice injected with either wild-type SC5314 or *ΔΔpfk1*. Fungal cells appear black in colour against the colored background of the kidney tissue. Bright field imaging was done using Axioplan 2 microscope at 40X magnification. Scale bar represents 50µm. **(H)** Schematic overview of N-acetyl cysteine (NAC) administration to mice. **(I)** 6mg/ml of NAC was dissolved in distilled water and administered orally to 6-8-week-old C57BL/6 mice before 72 hours of *C. albicans* infection. Mice administered with normal distilled water were used as controls. After 72 hours of treatment, mice were infected intravenously *via* lateral tail vein injection with 1 × 10^7 CFU of SC5314 or 1 × 10^7 CFU of *ΔΔpfk1*. To determine survival rate, mice were monitored for 21 days post-infection for clinical signs of illness or mortality. The results are representative of six mice per condition. Survival percentage were statistically evaluated by the Log-rank Mantel-Cox test, ***(P<0.001) and **(P<0.01) (n=6). **(J)** Histopathological analysis of kidney sections using GMS staining was done for kidneys isolated from mice administrated with normal distilled water or NAC containing distilled water and injected with *ΔΔpfk1*. Fungal cells appear black in colour against the colored background of the kidney tissue. Bright field imaging was done using Axioplan 2 microscope at 40X magnification. Scale bar represents 50µm. This figure was created using *Biorender.com*

Given that the perturbation of glycolysis under nitrogen-limiting conditions, using 2DG exhibited a strong inhibitory effect on hyphal differentiation in *C. albicans* (Fig. 4B), similar to what we observed in *S. cerevisiae*, we wanted to determine if the expression of genes involved in the *de novo* biosynthesis of sulfur-containing amino acids are affected when glycolysis is perturbed in *C. albicans*, under nitrogen-limiting conditions (similar to what we observed in *S. cerevisiae*). In order to determine this, we spotted *C. albicans* wild-type SC5314 on SLAD, in the presence and absence of 2DG and cells from colonies were isolated after ~4 days for RNA isolation. We then performed comparative RT-qPCR to check for the relative expression of genes involved in *de novo* biosynthesis of sulfur-containing amino acids including *met32* (DNA-binding transcription factor), *met3* (ATP sulfurylase), *met5* (*ecm17*) (Sulfite reductase beta subunit), *met10* (Sulfite reductase), and *met17* (*met15*) (O-acetyl homoserine-O-acetyl serine sulfhydrylase) (Chebaro et al., 2017; Li et al., 2013; Shrivastava et al., 2021), in cells isolated from colonies grown on SLAD containing 2DG compared to cells isolated from colonies grown on SLAD without 2DG (Fig. 4C). Our data clearly shows that all the aforesaid genes involved in *de novo* biosynthesis of sulfur-containing amino acids pathway including *met32*, *met3*, *met5* (*ecm17*), *met10* and *met17* (*met15*) were significantly downregulated in the presence of 2DG (Fig. 4D) suggesting that perturbation of glycolysis negatively impacts sulfur metabolism in *C. albicans*, under nitrogen-limiting conditions. To determine if the significant reduction in hyphal differentiation that we observed in the presence of 2DG can be attributed to the reduced expression of genes involved in the *de novo* biosynthesis of sulfur-containing amino acids, we performed sulfur add-back assays similar to what was done in *S. cerevisiae* (Fig. 3A), wherein *C. albicans* wild-type SC5314 was spotted on SLAD containing 2DG in the presence and absence of cysteine or methionine, allowed to grow for 7 days, and cells were isolated and imaged using bright-field microscopy. The addition of cysteine or methionine significantly rescued hyphal differentiation defects in the presence of 2DG (Fig. 4E and F). We then isolated cells from these colonies and quantified the percentage of hyphal cells. Corroborating our microscopy data, the percentage of hyphal cells in colonies spotted on SLAD containing 2DG with cysteine or methionine were significantly higher compared to the percentage of hyphal cells in colonies spotted on SLAD with 2DG alone (Fig. 4E and F).

Our previous results clearly demonstrate that glycolysis and glycolysis-dependent sulfur metabolism is critical for fungal morphogenesis of *C. albicans*, under nitrogen-limiting conditions. In order to corroborate these observations by using pertinent genetic knockout strains that are compromised in performing glycolysis, we generated a double knockout strain lacking the gene that encodes for the glycolytic enzyme, phosphofructokinase-1 (Pfk1) using the SAT-Flipper method (Reuß et al., 2004). The deletion of *pfk1* was confirmed through whole genome sequencing (Suppl.Fig. 5A). We then tested the ability of this deletion strain to undergo hyphal differentiation under nitrogen-limiting conditions. Briefly, *C. albicans* wild-type SC5314 and *ΔΔpfk1* were spotted on SLAD, allowed to grow for 7 days, and cells were isolated and imaged using bright-field microscopy. Our result shows that the *ΔΔpfk1* strain exhibits a significant reduction in hyphal differentiation compared to *C. albicans* wild-type SC5314, under nitrogen-limiting conditions (Fig. 4G). We then isolated cells from these colonies and quantified the percentage of hyphal cells in these colonies. Corroborating our microscopy data, the percentage of hyphal cells in *C. albicans* wild-type SC5314 colonies spotted on SLAD were significantly higher compared to the percentage of hyphal cells in *ΔΔpfk1* colonies (Fig. 4G). In order to rule out whether the reduction in hyphal differentiation in this mutant, is due to a general growth defect, we performed growth curve analysis wherein overnight grown cultures of deletion strain (*ΔΔpfk1)* along with wild-type strain (SC5314) were diluted to OD_{600}=0.01 in fresh SLAD and allowed to grow at 30 °C for 24 hours. OD_{600} was recorded at 3-h intervals. Our data suggests that the reduction in hyphal differentiation is not due to general growth defect as the mutant was able to grow similar to the wild-type strain in SLAD (Fig. 4H). We next asked if the exogenous addition of cAMP rescues the hyphal differentiation defect exhibited by glycolysis defective strain, *ΔΔpfk1*. In order to do this, we performed similar cAMP add-back assays wherein *C. albicans* wild-type SC5314 along with *ΔΔpfk1* strain were spotted on SLAD in the presence and absence of 5mM cAMP. Corroborating our previous data, the exogenous addition of cAMP failed to rescue hyphal differentiation defect caused by the perturbation of glycolysis *via* the deletion of *pfk1* (Suppl.Fig. 5C).

In order to determine if the expression of genes involved in the *de novo* biosynthesis of sulfur-containing amino acids are affected in *ΔΔpfk1*, under nitrogen-limiting conditions (similar to what we observed in the presence of 2DG (Fig. 4D)), we spotted *C. albicans* wild-type SC5314 and *ΔΔpfk1* on SLAD and cells from colonies were isolated after ~4 days, for RNA isolation. We then performed comparative RT-qPCR to check for the relative expression of genes involved in *de novo* biosynthesis of sulfur-containing amino acids including *met32* (DNA-binding transcription factor), *met3* (ATP sulfurylase), *met5* (*ecm17*) (Sulfite reductase beta subunit), *met10* (Sulfite reductase) and *met17* (*met15*) (O-acetyl homoserine-O-acetyl serine sulfhydrylase) (Chebaro et al., 2017; Li et al., 2013; Shrivastava et al., 2021)(Fig. 4C). Our data clearly shows that all the aforesaid genes involved in *de novo* biosynthesis of sulfur-containing amino acids pathway including *met32*, *met3*, *met5* (*ecm17*), *met10* and *met17* (*met15*) were significantly downregulated in *ΔΔpfk1* compared to the wild-type (Fig. 4I) suggesting that the genetic perturbation of glycolysis negatively impacts sulfur metabolism in *C. albicans*, under nitrogen-limiting conditions.

Our RT-qPCR results showed that the, deletion of *pfk1*, negatively influences sulfur metabolism and this in turn might be causal for the reduced hyphal differentiation observed in these conditions. Given this, we wanted to determine if the hyphal differentiation defect exhibited by *ΔΔpfk1* can be rescued by the addition of sulfur-containing amino acids. In order to do this, we performed sulfur add-back assays wherein *ΔΔpfk1* strain was spotted on SLAD in the presence and absence of cysteine or methionine, allowed to grow for 7 days and imaged using bright field microscopy. Interestingly, our data shows that cysteine or methionine were able to rescue the hyphal differentiation defect exhibited by the *ΔΔpfk1* strain (Fig. 4J and K). We then isolated cells from these colonies and quantified the percentage of hyphal cells. Corroborating our microscopy data, the percentage of hyphal cells in *ΔΔpfk1* colonies spotted on SLAD containing cysteine or methionine were significantly higher compared to the percentage of hyphal cells in *ΔΔpfk1* colonies spotted on SLAD alone (Fig. 4J and K). Overall, our data suggests that glycolysis-mediated regulation of sulfur metabolism is critical for hyphal differentiation of *C. albicans*, under nitrogen-limiting conditions highlighting the remarkable similarities with respect to the regulation of fungal morphogenesis, between *S. cerevisiae* and *C. albicans*.

Perturbation of Glycolysis Leads to Attenuated Fungal Virulence in *C. albicans* which is Rescued by Sulfur Supplementation

Several studies have established the importance of fungal morphogenesis as a key virulence strategy used by *C. albicans* to establish successful infections in the host (Gow et al., 2012; Moyes et al., 2016; Wilson et al., 2016). Based on our previous observations that perturbation of glycolysis, either through 2DG treatment or deletion of *pfk1* (*ΔΔpfk1*), significantly attenuates hyphal differentiation, we hypothesized that this observed phenotype could be a consequence of transcriptional downregulation of genes associated with hyphal differentiation and virulence. In order to test this hypothesis, we spotted *C. albicans* wild-type SC5314 on SLAD, in the presence and absence of 2DG or wild-type along with *ΔΔpfk1* on SLAD. Cells from colonies were isolated after ~4 days for RNA isolation. We then performed comparative RT-qPCR to check for the relative expression of genes involved in hyphal differentiation and virulence including *als3* (Cell wall adhesin), *ece1* (Candidalysin), *hwp1* (Hyphal cell wall protein), *hyr1* (GPI-anchored hyphal cell wall protein), *ihd1* (GPI-anchored protein), *rbt1* (Cell wall protein) and *sap6* (Secreted aspartyl protease) (Bailey et al., 1996; Deorukhkar, 2017; Liu and Filler, 2011; Martin et al., 2013; Monniot et al., 2013; Moyes et al., 2016; Richardson et al., 2018; Sharkey et al., 1999) (Fig. 5A). Our data clearly shows that all the aforesaid genes involved in hyphal differentiation and virulence including *als3, ece1, hwp1, hyr1, ihd1, rbt1* and *sap6* were significantly downregulated in both 2DG treated cells and in *ΔΔpfk1* suggesting that perturbation of glycolysis negatively affects the expression of genes involved in hyphal differentiation and virulence in *C. albicans*, under nitrogen-limiting conditions (Fig. 5B).

Given that *ΔΔpfk1*, displayed a significant attenuation in its ability to undergo hyphal differentiation and it had significantly reduced expression of genes associated with hyphal differentiation and virulence, we wanted to determine if this deletion strain was compromised in its ability to infect the host. In order to do this, we took a two-pronged approach wherein, we first tested the ability of the *ΔΔpfk1* strain to survive the harsh intracellular environment of the macrophages which *C. albicans* encounters during various stages of host infection (Jiménez-López and Lorenz, 2013; May and Casadevall, 2018; Uwamahoro et al., 2014). We performed an *in vitro* microbial survival assay wherein *RAW 264.7* macrophages were incubated either with *C. albicans* wild-type SC5314 or the *ΔΔpfk1* strain for one hour following which macrophages were lysed and fungi were serially diluted and plated to determine the viable population of the *C. albicans* wild-type SC5314 and the *ΔΔpfk1* strain (Fig. 5C). We also calculated the percentage of the wild-type or mutant strain that survived the incubation with the macrophages. Our results demonstrate that the viable population of *ΔΔpfk1* strain was significantly lower compared to the wild-type strain, after incubation with the macrophages (Fig. 5C). Similarly, the percentage survival of the *ΔΔpfk1* strain was significantly lower than the wild-type SC5314 (Fig. 5C). Overall, this data indicates that the *ΔΔpfk1* strain is compromised with respect to macrophage survival.

Given that the *ΔΔpfk1* strain was significantly attenuated in the macrophage survival assay, we next wanted to test the ability of this deletion strain to establish systemic infection in a host. In order to do this, we challenged healthy wild-type C57BL/6 mice intravenously with a dose of 1×10^7 CFU (lethal dose (Hirayama et al., 2020)) of *C. albicans* wild-type SC5314 or *ΔΔpfk1* strain *via* lateral tail vein injection. This is an established *murine* model of systemic candidiasis in the field and faithfully mimics the infection that occurs in humans (Segal and Frenkel, 2018). We then monitored survival of the infected animals for 21 days. Mice challenged with *ΔΔpfk1* strain had significantly increased survival compared to mice challenged with *C. albicans* wild-type SC5314 (Fig. 5D). *C. albicans* exhibits strong tropism towards kidneys during *murine* infections (Lionakis et al., 2011; Pappas et al., 2018) and in order to measure the fungal burden within the host, kidneys from C57BL/6 mice infected with both *C. albicans* wild-type SC5314 and *ΔΔpfk1* were harvested ~2 days post infection (when the mice injected with wild-type *C. albicans* SC5314 becomes moribund), homogenized and plated to measure the fungal burden by enumerating the colony forming units (CFU) (Fig. 5E). Our data indicates that kidneys isolated from mice challenged with the *ΔΔpfk1* strain had significantly lesser CFU compared to kidneys isolated from mice challenged with the *C. albicans* wild-type SC5314 (Fig. 5F). We also performed histological analysis on the infected kidneys using GMS staining (Grocott Methenamine Silver staining), to visualize the fungi (stained black) in the tissue. Kidneys isolated from mice challenged with the *C. albicans* wild-type SC5314 had higher amounts of fungi compared to kidneys isolated from mice challenged with the *ΔΔpfk1* strain (Fig. 5G). Overall, our data indicates that the perturbation of glycolysis significantly attenuates *C. albicans* virulence.

Our previous *in vitro* results clearly demonstrated that exogenous supplementation of cysteine and methionine significantly rescued the hyphal differentiation defect exhibited by *ΔΔpfk1* strain. Based on this, we wanted to check whether sulfur supplementation could also rescue the virulence defects exhibited by *ΔΔpfk1*, in a *murine* model of systemic candidiasis. Given that N-acetyl cysteine (NAC) is a commonly used exogenous sulfur source for *in vivo* studies (Shee et al., 2022), we supplemented distilled water with NAC (6mg/ml concentration) and this was administered to mice orally 72 hours prior to infection. After 72 hours of pre-treatment with NAC, we challenged healthy wild-type C57BL/6 mice intravenously with a dose of 1×10^7 CFU of wild-type SC5314 or *ΔΔpfk1* strain *via* lateral tail vein injection (Fig. 5H). We then monitored survival of the infected animals for 21 days. Remarkably, mice administered with NAC containing distilled water had significantly reduced survival compared to mice administrated with normal distilled water (Fig. 5I). We also performed histological analysis on the infected kidneys using GMS staining, to visualize the fungi in the tissue. Kidneys isolated from mice administered with NAC containing distilled water and infected with *ΔΔpfk1* had significantly higher amounts of fungi compared to mice administrated with normal distilled water (Fig. 5J). This clearly demonstrates that exogenous sulfur supplementation to the host *via* NAC administration, reverses the attenuated virulence exhibited by the *ΔΔpfk1* strain.

Discussion

Our findings provide compelling evidence for a conserved metabolic network that intricately links central carbon metabolism (particularly glycolysis), sulfur amino acid biosynthesis, and fungal morphogenesis in both *Saccharomyces cerevisiae* and *Candida albicans*. While nitrogen limitation is a well-established and essential trigger for the yeast to pseudohyphae transition in *S. cerevisiae*, the specific influence of carbon sources on this process has not been completely understood. Studies have shown that fermentable carbon sources are critical for inducing pseudohyphal differentiation under nitrogen-limiting conditions (Van De Velde and Thevelein, 2008). Conversely, when these fermentable sugars are replaced with non-fermentable carbon sources, a significant reduction in pseudohyphal differentiation is observed (Strudwick et al., 2010). These findings collectively suggest that the presence of fermentable carbon sources, and by extension their metabolism, is a critical requirement for inducing fungal morphogenesis in *S. cerevisiae*. Glucose in the extracellular environment can be sensed by *S. cerevisiae* using multiple proteins. One such protein is the well characterized G-protein coupled receptor (GPCR), Gpr1. Binding of glucose to Gpr1 activates a downstream signalling cascade, resulting in the production of cAMP which activates Protein Kinase A (PKA) (Kraakman et al., 1999; Yun et al., 1997). This glucose-dependent cAMP-PKA pathway has been shown to be essential for pseudohyphal differentiation under nitrogen-limiting conditions (Lorenz et al., 2000). Additionally, a key glycolytic intermediate, fructose-1,6-bisphosphate (FBP), is also known to activate the cAMP-PKA pathway *via* activation of Ras proteins, further linking glucose metabolism to this crucial signalling cascade (Peeters et al., 2017). However, the role of glucose as a metabolite in the context of pseudohyphal differentiation is not known. Building upon previous data that fermentable carbon sources that are metabolized *via* glycolysis are able to induce fungal morphogenesis, we asked whether the ability of these cells to undergo active glycolysis is critical for fungal morphogenesis. We took a two-pronged approach which involved pharmacological or genetic perturbation of glycolysis, to address this question. Our findings clearly demonstrated that active glycolysis is essential for fungal morphogenesis under nitrogen-limiting conditions. To understand whether the pseudohyphal differentiation defects we observed due to glycolysis perturbation were solely due to the inability of glucose to activate glucose-dependent cAMP-PKA pathway, we performed cAMP add-back assays and interestingly, exogenous supplementation of cAMP failed to rescue pseudohyphal differentiation defects caused by the perturbation of glycolysis (through inhibitors (2DG and NaCi) or through genetic knockout strains (*ΔΔpfk1* and *ΔΔadh1*)). Our results clearly demonstrate that the ability of *S. cerevisiae* to efficiently metabolize glucose *via* glycolysis is critical for pseudohyphal differentiation under nitrogen-limiting conditions, in a cAMP-PKA independent manner. This implies that glycolysis may be parallelly regulating other cellular mechanisms essential for this morphological transition. It is plausible that the glycolytic flux, or a yet to be identified downstream intermediate, acts as a distinct metabolic signal that is parallelly important for pseudohyphal differentiation along with the cAMP-PKA pathway. This uncovers a previously unrecognized layer of regulatory complexity in the interplay between central carbon metabolism and fungal morphogenesis.

To gain deeper insights underlying the glycolysis-mediated regulation of fungal morphogenesis in a cAMP-PKA independent manner, we performed comparative transcriptomics on *S. cerevisiae* wild-type cells grown under nitrogen-limiting conditions with and without the glycolysis inhibitor, 2DG (which had the strongest inhibitory effect on pseudohyphal differentiation). As anticipated, genes involved in amino acid biosynthesis and transport were upregulated in the 2DG treated cells, consistent with a feedback response that cells exhibits when glycolysis is perturbed under nitrogen-limiting conditions, as various intermediates of glycolysis are known to serve as precursors for the biosynthesis of several amino acids under these conditions (Dikicioglu et al., 2011; Martíez-force and Benítez, 1992). Remarkably, we observed a striking and unexpected downregulation of multiple genes specifically involved in the biosynthesis and transport of sulfur-containing amino acids, cysteine and methionine, in the 2DG treated cells at both early and late time points of pseudohyphal development. This transcriptional downregulation was further validated at the protein level for key proteins involved in the sulfur assimilation pathway (Met4, Met32, Met16, Met10, Cys4 and Cys3), clearly demonstrating the dependence of sulfur metabolism on active glycolysis during fungal morphogenesis. This unexpected link between central carbon metabolism and sulfur metabolism is not restricted to conditions that induce fungal morphogenesis *i.e.* in surface-attached colonies grown in nitrogen-limiting conditions since even in liquid cultures that have limiting levels of nitrogen, wherein *S. cerevisiae* cells do not undergo pseudohyphal differentiation (Gancedo, 2001; Gimeno et al., 1992), perturbation of glycolysis with 2DG results in significantly reduced expression of proteins involved in the sulfur assimilation pathway (Met32, Met16, Met10 and Cys3).

Given that multiple genes that encode for proteins involved in *de novo* biosynthesis of sulfur-containing amino acids were downregulated in the presence of 2DG, which strongly inhibits fungal morphogenesis, we wanted to explore the possibility that perturbation of sulfur metabolism is causal for the attenuation of fungal morphogenesis in the presence of 2DG. We tested the functional relevance of this glycolysis-dependent regulation of sulfur metabolism in fungal morphogenesis directly through exogenous supplementation of sulfur sources. Interestingly, cysteine specifically rescued the pseudohyphal differentiation defect exhibited by *S. cerevisiae* in the presence of 2DG. However, methionine was unable to rescue the pseudohyphal differentiation defect, in the presence of 2DG even at various different concentrations (50µM, 100 µM, 200µM and 500µM). The observed differential rescue effects of cysteine and methionine could be due to the distinct amino acid transport systems used by *S. cerevisiae* to transport these amino acids. *S. cerevisiae* primarily uses multiple, low-affinity permeases (Gap1, Bap2, Bap3, Tat1, Tat2, Agp1, Gnp1, Yct1) for cysteine transport, while relying on a limited set of high-affinity transporters (like Mup1) for methionine transport, with the added complexity that its methionine transporters can also transport cysteine (Düring-Olsen et al., 1999; Huang et al., 2017; Kosugi et al., 2001; Menant et al., 2006). Hence, it is likely that cysteine uptake could be happening at a higher efficiency in *S. cerevisiae* compared to methionine uptake. Therefore, to achieve a comparable functional rescue by exogenous supplementation of methionine, it is necessary to use a higher concentration of methionine. When we performed our rescue experiments using higher concentrations of methionine, we did not see any rescue of pseudohyphal differentiation in the presence of 2DG and in fact we noticed that, at higher concentrations of methionine, the wild-type strain failed to undergo pseudohyphal differentiation even in the absence of 2DG. This is likely due to the fact that increasing the methionine concentration raises the overall nitrogen content of the medium, thereby making the medium less nitrogen-starved. This presents a major experimental constraint, as pseudohyphal differentiation is strictly dependent on nitrogen limitation, and the elevated nitrogen resulting from the higher methionine concentration can inhibit pseudohyphal differentiation. Similarly, gene expression analysis using RT-qPCR revealed a significant downregulation of multiple genes involved in the biosynthesis of sulfur-containing amino acids (*met32*, *met3, met5* and *met17)* in *ΔΔpfk1* and *ΔΔadh1* strains and the pseudohyphal differentiation defects observed in these strains were completely rescued by the addition of cysteine or methionine. These findings strongly implicate that *de novo* biosynthesis of sulfur-containing amino acids is a critical downstream effector of glycolytic activity in promoting pseudohyphal differentiation under nitrogen limitation. To directly assess the role of sulfur metabolism in fungal morphogenesis, we generated and characterized the *met32* deletion strain. This particular strain was critical for our investigation because, unlike other *MET* gene deletions, *met32* deletion does not result in auxotrophy for methionine or cysteine. This unique characteristic is attributed to the functional redundancy provided by its paralog, *met31* (Blaiseau et al., 1997). The deletion of *met32* resulted in substantial defects in pseudohyphal differentiation under nitrogen-limiting conditions, directly implicating the role of sulfur metabolism in regulating fungal morphogenesis. Our sulfur add-back experiments demonstrated that cysteine or methionine rescued the morphogenetic defects exhibited by the *met32* knockout strain, further solidifying the direct involvement of sulfur metabolism in this morphogenetic switching process. How active sulfur metabolism contributes towards fungal morphogenesis is not yet understood but given its critical requirement during this phenomenon, it would be an important question to explore and will further our overall understanding of this important biological phenomenon.

It is interesting to note that the expression of Met4, the principal transcriptional regulator of the *de novo* biosynthesis of sulfur-containing amino acids in *S. cerevisiae* along with Met32 one of its cognate DNA-binding cofactors (Blaiseau et al., 1997), are significantly downregulated when glycolysis is perturbed. Given the obligate requirement of Met4 interaction with DNA-binding proteins such as Met28, Met32, or Met31, to elicit transcriptional activation of the genes involved in the *de novo* biosynthesis of sulfur-containing amino acids (Blaiseau and Thomas, 1998), the observed downregulation in both Met4 and Met32 levels likely compromises the overall biosynthesis of sulfur-containing amino acids. This concurrent downregulation could explain the attenuated expression of downstream genes involved in the synthesis of methionine and cysteine. This suggests a novel regulatory mechanism wherein the expression of Met4/Met32 is coupled to glycolytic activity, under nitrogen-limiting conditions. Given that sulfur metabolism is coupled to glycolysis in *C. albicans* as well and contributes towards fungal morphogenesis, it is likely that this conserved regulatory axis might be critical for fungal morphogenesis in a broad range of fungal species. Interestingly, Met4 regulation is directly dependent on Met30 (Rouillon et al., 2000). Met30 functions as a molecular switch for Met4, the master regulator of the *MET* gene network. Under sulfur replete conditions, Met30 ensures that Met4 remains inactive through ubiquitination and proteasomal degradation. Conversely, a reduction in sulfur availability diminishes Met30 activity, thereby de-repressing Met4 and allowing it to induce the expression of genes necessary for sulfur amino acid biosynthesis (Rouillon et al., 2000; Smothers et al., 2000; Thomas and Surdin-Kerjan, 1997). Given this regulatory mechanism, we investigated the expression of proteins involved in the sulfur assimilation pathway (Met4, Met32, Met16 and Cys3), in a viable *Δmet30* deletion strain, under conditions where glycolysis was perturbed using 2DG. We observed no difference in protein levels of Met4, Met32, Met16 and Cys3, between 2DG treated and untreated cells. This indicates that, in the heterozygous deletion strain of *met30* (*Δmet30*), Met4 is not degraded even when glycolysis is perturbed, supporting the hypothesis that perturbing active glycolysis with inhibitors like 2DG increases Met30 activity, which in turn leads to the increased degradation of Met4, which would then negatively affect *MET* gene transcription subsequently attenuating pseudohyphal differentiation. However, a surprising finding emerged when we examined Met30 levels in the presence of 2DG. There was no increase in the levels of Met30 in 2DG treated cells compared to untreated cells. This suggests that glycolysis influences Met30 activity through a post-translational mechanism, rather than by altering its protein abundance. Studies have shown that under cadmium stress, Met4 activity is upregulated by reducing the activity of Met30 through post-translational events, specifically by inducing the dissociation of Met30 from the core SCF (Skp1/Cullin1/F-box) complex without affecting Met30 protein levels (Lauinger et al., 2024; Yen et al., 2005). This well-established paradigm provides a compelling framework for our observation, suggesting that active glycolytic flux, similar to cadmium stress, might rapidly modulate the activity of Met30 by decreasing its association with the SCF complex to control Met4 activity, rather than through changes in protein synthesis or degradation. Future studies will be needed to delineate the precise post-translational modifications or regulatory interactions that modulate Met30 activity in response to glycolytic flux.

Intriguingly, our work extended these findings to the pathogenic fungus *C. albicans*, revealing a conserved role for glycolysis-mediated sulfur metabolism in hyphal differentiation, a key virulence attribute in this species (Gow et al., 2012; Moyes et al., 2016; Wilson et al., 2016). Similar to our observations in *S. cerevisiae*, *C. albicans* exhibited glycolysis-dependent hyphal formation under nitrogen limitation in a cAMP-PKA pathway independent manner. Perturbation of glycolysis (either by 2DG addition or the deletion of *pfk1*) resulted in the significant downregulation of genes involved in *de novo* biosynthesis of sulfur-containing amino acids (*met32*, *met3*, *met5* (*ecm17*), *met10* and *met17* (*met15*)) and the hyphal differentiation defects that occur in response to glycolysis perturbation could be rescued by the exogenous supplementation of cysteine or methionine, wherein the exogenous addition of cAMP failed to rescue the hyphal defects caused by the addition of 2DG. These results clearly demonstrate that *de novo* biosynthesis of sulfur-containing amino acids is a critical downstream effector of glycolytic activity in promoting hyphal differentiation under nitrogen-limiting conditions, in *C. albicans* as well. Given that central carbon metabolic pathways like glycolysis and sulfur assimilation pathway are broadly conserved metabolic processes across a plethora of fungal species, it is possible that this novel regulatory axis might play a crucial role in modulating fungal morphogenesis in multiple fungal species and further studies are warranted to explore this. Finally, given the established link between hyphal differentiation and *C. albicans* virulence, we explored the impact of glycolytic perturbation on the pathogenicity of this fungus. Interestingly, RT-qPCR analysis revealed a significant downregulation of genes involved in hyphal differentiation and virulence (*als3, ece1, hwp1, hyr1, ihd1, rbt1* and *sap6*) upon 2DG treatment and in the *ΔΔpfk1* strain. Interestingly, deletions of sugar kinases or transcription factors regulating the expression of glycolysis genes or *adh1* in *C. albicans* leads to filamentation defects and downregulation of filamentation/virulence-specific genes (Askew et al., 2009; Laurian et al., 2019; Song et al., 2019). Our data, showing transcriptional downregulation of hyphal differentiation and virulence genes following glycolytic perturbation *via* the deletion of *pfk1*, strongly corroborate these previous findings. This collectively positions glycolysis as a central metabolic nexus, rather than merely a catabolic process, profoundly influencing the pathogenic ability of *C. albicans* by intricately linking cellular metabolism, morphogenetic transitions, and the expression of its virulence factors. Furthermore, the *ΔΔpfk1* strain displayed significantly reduced survival within *murine* macrophages and attenuated virulence in a *murine* model of systemic candidiasis, as evidenced by increased host survival, lower kidney fungal burden, and reduced tissue colonization. Remarkably, exogenous sulfur supplementation (N-acetyl cysteine (NAC)) to mice rescued the attenuated virulence exhibited by *ΔΔpfk1*. It is important to note that, in the context of certain bacterial pathogens, NAC has been reported to augment cellular respiration, subsequently increasing Reactive Oxygen Species (ROS) generation, which contributes to pathogen clearance (Shee et al., 2022). Interestingly, in our study, NAC supplementation to the mice was given prior to the infection and maintained continuously throughout the duration of the experiment. This continuous supply of NAC likely contributes to the rescue of virulence defects exhibited by the *ΔΔpfk1* strain. Essentially, NAC likely allows the mutant to fully activate its essential virulence strategies (including morphological switching), to cause a successful infection in the host. These findings underscore the critical role of glycolysis, and by extension, glycolysis-dependent sulfur metabolism, in the virulence of *C. albicans*.

In conclusion, our study unveils a conserved metabolic network that orchestrates fungal morphogenesis in response to nutrient availability. We demonstrate that active glycolysis, fueled by fermentable carbon sources in a cAMP-PKA independent manner, plays a crucial role in regulating the *de novo* biosynthesis of sulfur-containing amino acids in Met30-dependent manner, which in turn are essential for the morphological transitions in both *S. cerevisiae* and *C. albicans* (Fig. 6). The attenuation of *C. albicans* virulence upon disruption of this pathway highlights the potential of targeting specific fungal metabolic networks as a novel antifungal strategy. Future research in our laboratory will focus on elucidating the precise molecular mechanisms by which glycolytic intermediates or downstream signals regulate the expression of sulfur metabolism pathway and how sulfur-containing amino acids contribute to the cellular processes underlying fungal morphogenesis.

Materials and Methods

Yeast Strains

The prototrophic diploid strains, ∑1278b or CEN.PK were used in all the experiments involving *S. cerevisiae*. The prototrophic diploid strain, SC5314 was exclusively used in all the experiments involving *C. albicans*. The yeast strains used in this study are listed in Suppl.Table 1.

Generation of *S. cerevisiae* Gene Knockout and Epitope-tagged Strains

A standard PCR-based technique was used to amplify resistance cassettes (G418, HygB, or NAT) with flanking sequences in order to carry out gene deletions. The target gene was then replaced by homologous recombination using the lithium acetate-based transformation (Schiestl and Gietz, 1989). Similarly, C-terminal epitope-tagged yeast strains were generated using a PCR-based technique. This involved amplifying resistance cassettes, flanked by sequences homologous to the C-terminal region of the target gene, along with the desired epitope tags. The targeted gene was then tagged at its C-terminus through homologous recombination, using the lithium acetate-based transformation method. Primers used for these experiments are listed in Suppl.Table 2.

Generation of *C. albicans* Gene Knockout Strains

The *C. albicans* strains used in this study are listed in Suppl.Table 1. The previously established SAT1-flipper method was used with some modifications, for the deletion of the desired gene (Reuß et al., 2004). To generate the *pfk1* deletion (*ΔΔpfk1*) strain, a DNA fragment carrying the flanking regions of *pfk1* (100 bps homology) and the *SAT1*-flipper cassette was transformed into *C. albicans* wild-type SC5314, using electroporation. Transformants were selected based on their nourseothricin (NAT) resistance. PCR confirmation was used to validate the transformants. The *SAT1*-flipper cassette was then excised from the *pfk1* locus by growing the cells in YPM medium (10g/l yeast extract, 20g/l peptone, and 2% (w/v) maltose) to induce expression of the *MAL2* promoter-regulated recombinase. To knock out the second allele of *pfk1*, the heterozygous *pfk1* deletion mutants (*Δpfk1/PFK1)* were used for the transformation wherein the DNA fragment carrying the flanking regions of *pfk1* (100 bps homology) and the *SAT1*-flipper cassette was transformed into *C. albicans Δpfk1/PFK1* SC5314. Transformants were selected based on their NAT resistance. PCR confirmation was used to validate the transformants. Primers used to generate or confirm the deletion strains are listed in Suppl.Table 2.

Media and Growth Conditions

YPD broth containing 10g/l yeast extract, 20g/l peptone and 2% (w/v) glucose was used for the overnight growth of the cultures. Nitrogen-limiting medium containing 1.7g/l yeast nitrogen base without amino acids or ammonium sulfate, 2% (w/v) glucose (SLAD), 50µM ammonium sulfate and 2% (w/v) agar which was extensively washed, to remove excess nitrogen (Gimeno et al., 1992) was used for all experiments involving pseudohyphal or hyphal differentiation. Overnight

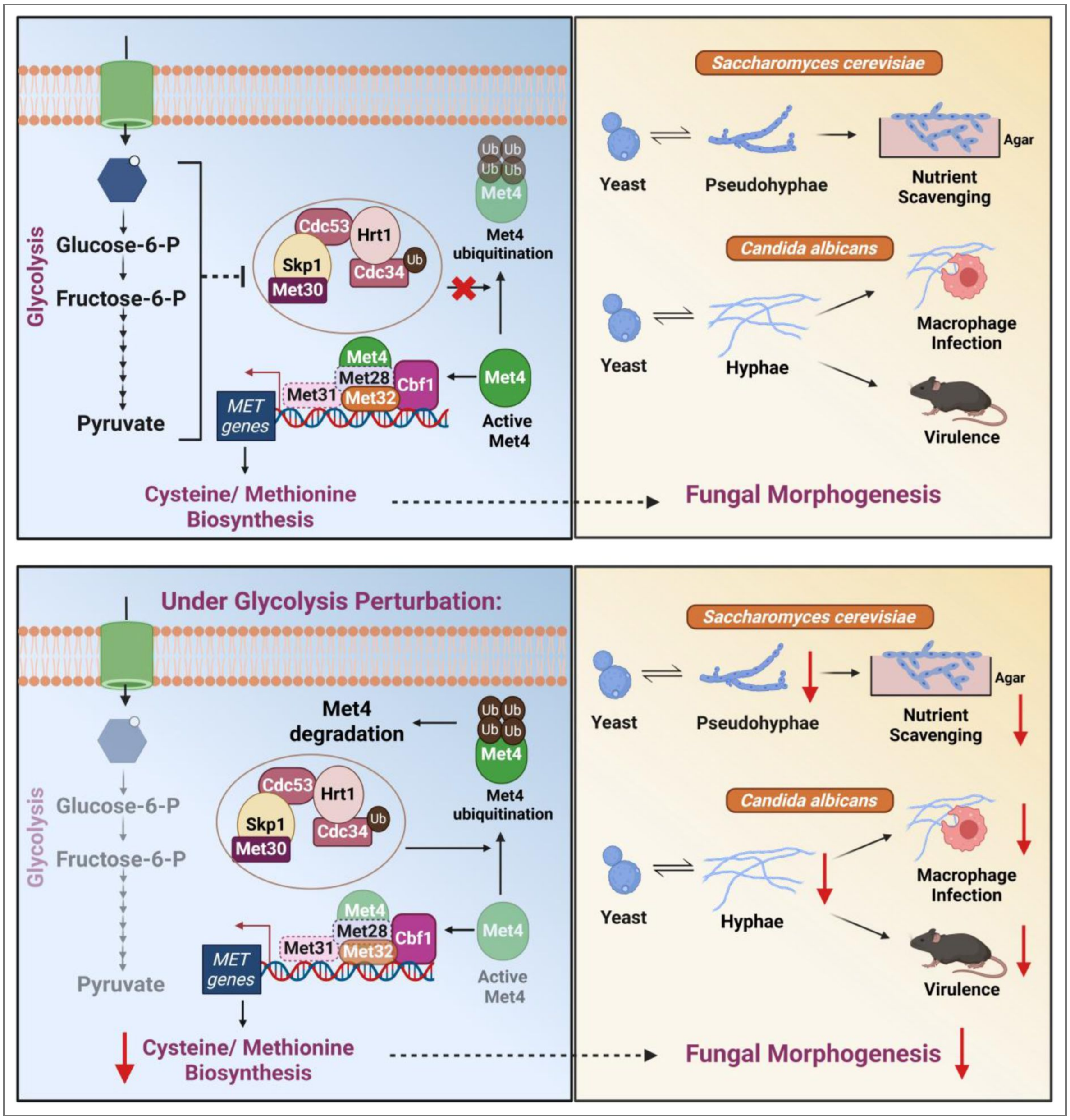


Figure 6. **Glycolysis-dependent Sulfur Metabolism Orchestrates Morphological Transitions and Virulence in Fungi.**

Active glycolysis plays a key regulatory role in the *de novo* biosynthesis of sulfur-containing amino acids by modulating the activity of SCF^{Met30} complex (SCF^{Met30} complex includes Met30, Skp1, Cdc53, Hrt1 and Cdc34) which in turn affects the expression of genes involved in this process (In *S. cerevisiae*, *MET* transcription complex includes Met4, Met28, Cbf1, Met31, and Met32, whereas in *C. albicans*, the characterized components of this complex include Met4, Cbf1, and Met32. Hence, Met28 and Met31, unique to the *MET* transcription complex of *S. cerevisiae* are represented with dotted borders). Glycolysis-dependent sulfur metabolism in turn, is critical for fungal morphogenesis in both species and the ability of *C. albicans* to survive within macrophages and cause systemic infection in a *murine* model of candidiasis. Perturbation of active glycolysis increases Met30 activity, which leads to the increased degradation of Met4, resulting in the reduced expression of genes involved in the *de novo* biosynthesis of sulfur-containing amino acids. This in turn attenuates fungal morphogenesis in both species and the ability of *C. albicans* to survive within macrophages and cause systemic infection in a *murine* model of candidiasis. This figure was created using *Biorender.com* cultures of various strains were grown in YPD broth and incubated at 30 °C with continuous shaking at 200 rotations per minute (RPM). Reagents and tools used in this study are mentioned in Suppl.Table 3.

Pseudohyphal Differentiation Assay

To induce pseudohyphal differentiation in *S. cerevisiae,* overnight cultures of various strains (listed in Suppl.Table 1) were spotted on nitrogen-limiting medium (SLA) containing 2% (w/v) glucose (SLAD). In order to check the effect of glycolysis inhibitors, 0.05% (w/v) concentration of 2-Deoxy-D-Glucose (2DG) and 0.2% (w/v) concentration of sodium citrate (NaCi), on pseudohyphal differentiation, wild-type was spotted on SLAD containing sub-inhibitory concentrations of 2DG (0.05% w/v) or NaCi (0.2% w/v). Plates were incubated for 10 days at 30 °C (Chandarlapaty and Errede, 1998; González et al., 2017). After 10 days, imaging was done using Olympus MVX10 stereo microscope. Cells from colonies were isolated and single cells were imaged using Zeiss Apotome microscope. Length/width ratio of individual cells was measured using ImageJ and percentage of pseudohyphal cells from total population was determined. Cells with a length/width ratio of 2 or more were considered as pseudohyphal cells as described previously (Schröder et al., 2000). Statistical analysis was done using unpaired t-test or one-way ANOVA test. Error bars represent SEM. All the experiments were performed independently, thrice.

Hyphal Differentiation Assay

To induce hyphal differentiation in *C. albicans,* overnight cultures of each strain (listed in Suppl.Table 1) were spotted on nitrogen-limiting medium (SLAD). To check the effect of glycolysis inhibitor, 2DG on hyphal differentiation, wild-type was spotted on SLAD containing sub-inhibitory concentration of 2DG (0.2% w/v). All the plates were incubated for 7 days at 37 °C (Sánchez-Martínez and Pérez-Martín, 2002; Song et al., 2019). After 7 days, cells from colonies were isolated and single cells were imaged using Zeiss Apotome microscope. Length/width ratio of individual cells was measured using ImageJ and percentage of hyphal cells from total population was determined. Cells with a length/width ratio of 4.5 or more were considered as hyphal cells as described previously (Su et al., 2018). Statistical analysis was done using unpaired t-test or one-way ANOVA test. Error bars represent SEM. All the experiments were performed independently, thrice.

Growth Curves

Growth curve analysis in the presence of glycolysis inhibitors (2DG or NaCi) was performed as follows. Overnight grown cultures of wild-type strains (∑1278b or CEN.PK or SC5314) were diluted to OD_{600}=0.01 in fresh SLAD medium with and without 2DG or NaCi and allowed to grow at 30 °C for 24 hours. OD_{600} was recorded at 3-h intervals. Growth curve analysis using various deletion strains were performed as follows. Overnight grown cultures of various deletion strains (listed in Suppl.Table 1) along with wild-type strains (∑1278b or CEN.PK or SC5314) were diluted to OD_{600}=0.01 in fresh SLAD and allowed to grow at 30 °C for 24 hours. OD_{600} was recorded at 3-h intervals. Graphs were prepared using GraphPad Prism. Error bars represent SEM. All the experiments were performed independently, thrice.

cAMP Add-back Assays

In order to perform cAMP add-back assays, overnight cultures of wild-type ∑1278b and various deletion strains of *S. cerevisiae* (listed in Suppl.Table 1) were spotted on SLAD or SLAD containing sub-inhibitory concentration of 2DG or NaCi, in the presence and absence of cAMP (1mM) (Lorenz, 1997). Plates were incubated for 10 days at 30 °C. After 10 days, imaging was done using Olympus MVX10 stereo microscope. Cells from colonies were isolated and single cells were imaged using Zeiss Apotome microscope. Length/width ratio of individual cells was measured using ImageJ and percentage of pseudohyphal cells from total population was determined. For hyphal differentiation assays in *C. albicans*, overnight cultures of wild-type SC5314 and *ΔΔpfk1* strains of *C. albicans* were spotted on SLAD or SLAD containing sub-inhibitory concentration of 2DG, in the presence and absence of cAMP (5mM) (Li et al., 2013). Plates were incubated for 7 days at 37 °C. After 7 days, cells from colonies were isolated and single cells were imaged using Zeiss Apotome microscope. Length/width ratio of individual cells was measured using ImageJ and percentage of hyphal cells from total population was determined. Statistical analysis was done using one-way ANOVA test. Error bars represent SEM. All the experiments were performed independently, thrice.

Sulfur Add-back Assays

In order to perform sulfur add-back assays, overnight cultures of various strains of *S. cerevisiae* or *C. albicans* (listed in Suppl.Table 1) were spotted on SLAD or SLAD containing sub-inhibitory concentration of 2DG, in the presence and absence of sulfur-containing compounds including cysteine, methionine. Plates were incubated for 10 days at 30 °C to induce pseudohyphal differentiation in *S. cerevisiae*. After 10 days, imaging was done using Olympus MVX10 stereo microscope. Cells from colonies were isolated and single cells were imaged using Zeiss Apotome microscope. Length/width ratio of individual cells was measured using ImageJ and percentage of pseudohyphal cells from total population was determined. For hyphal differentiation assays in *C. albicans*, plates were incubated for 7 days at 37 °C. After 7 days, cells from colonies were isolated and single cells were imaged using Zeiss Apotome microscope. Length/width ratio of individual cells was measured using ImageJ and percentage of hyphal cells from total population was determined. Statistical analysis was done using unpaired t-test or one-way ANOVA test. Error bars represent SEM. All the experiments were performed independently, thrice.

Whole genome sequencing

Sample Collection

Strains of *S. cerevisiae* (*ΔΔpfk1* or *ΔΔadh1* along with wild-type strain ∑1278b) or *C. albicans* (*ΔΔpfk1* along with wild-type strain SC5314) were grown in YPD medium at 30 °C for overnight.

Genomic DNA Extraction

Genomic DNA from the samples was extracted using phenol-chloroform method. Following RNAse treatment to eliminate any contaminating RNA from the sample, the extracted DNA was submitted to the CCMB next-generation sequencing facility for whole genome sequencing.

Whole genome sequencing Analysis

Raw sequencing reads were obtained in FASTQ format. The reference genome and annotation files for the *S. cerevisiae* strain ∑1278b were downloaded from the Saccharomyces Genome Database (SGD) and reference genome and annotation files for the *C. albicans* strain SC5314 were downloaded from the NCBI (National Center for Biotechnology Information). Quality assessment of the raw reads was performed using FastQC (version 0.12.1). Adapter sequences were trimmed using Cutadapt (version 4.6). The reference genome was indexed with Hisat2 (version 2.2.1), and Samtools (version 1.20) was used to filter out multi-mapped reads and convert SAM files to BAM format. Visualisation of reads was done using IGV software (version 2.16.2).

RNA-Sequencing

Sample Collection

Overnight culture of *S. cereviae* wild-type ∑1278b was spotted on nitrogen-limiting media containing 2% (w/v) glucose (SLAD) with and without sub-inhibitory concentration of 2DG. The plates were incubated at 30 °C and colonies were isolated on day 5 (D-5) and day 10 (D-10).

RNA-Extraction

Total RNA from the sample was extracted using the protocol mentioned in the Ribopure RNA purification kit (yeast) (cat. #AM1926). Following a DNAse treatment to eliminate any genomic DNA from the sample, the extracted RNA was submitted to the CCMB next-generation sequencing facility for RNA-Seqencing. The NovaSeq 6000 equipment was used to sequence transcriptomes.

RNA-Sequencing Analysis

Raw sequencing reads were obtained in FASTQ format. The reference genome and annotation files for the *S. cerevisiae* strain ∑1278b were downloaded from the Saccharomyces Genome Database (SGD). Quality assessment of the raw reads was performed using FastQC (version 0.12.1). Adapter sequences were trimmed using Cutadapt (version 4.6). The reference genome was indexed with Hisat2 (version 2.2.1), and Samtools (version 1.20) was used to filter out multi-mapped reads and convert SAM files to BAM format. Read counting for uniquely mapped reads was done with FeatureCounts, using gene annotation data from the reference *S. cerevisiae* ∑1278b genome. Normalization and differential gene expression analysis were performed using R (version 4.3.2). Genes with a $\log_2$ fold change of ≥1 were considered to be upregulated, indicating at least a doubling in expression under the given condition, while those with a $\log_2$ fold change of ≤ −1 were considered to be downregulated.

Protein Isolation

For Western blotting experiments, in order to isolate proteins from liquid cultures, overnight yeast cultures were back-diluted in 10 ml of liquid SLAD in the presence and absence of 2DG. The yeast cells were grown till an OD_{600} of ~1 and cells were harvested by centrifugation at 3000 rpm for 10 minutes at room temperature (RT). To isolate proteins from colonies, overnight cultures of pertinent strains were spotted on SLAD with and without sub-inhibitory concentration of 2DG (0.05% (w/v)) and colonies were isolated and resuspended in 1ml of 1X PBS (Phosphate Buffered Saline) after 5 days (D-5). Cells were harvested by centrifugation at 14000 rpm for 10 minutes at room temperature. Pelleted cells (from liquid cultures and colonies) were resuspended in 300µl of ice cold 10% (w/v) TCA solution and disrupted using bead-beating by addition of 150µl of glass beads at 4°C. Supernatant was transferred to another tube. Next, proteins were pelleted by centrifugation at 14,000 rpm for 10 minutes at 4 °C and supernatant was discarded. Each pellet was resuspended in 400µl of SDS/glycerol buffer (7.3% (w/v) SDS, 29% (v/v) glycerol and 83.3mM tris-base) and boiled at 95 °C for 10 minutes with occasional vortexing followed by centrifugation at 14,000 rpm for 10 minutes at room temperature. Supernatant was transferred to a fresh tube and used for protein estimation using BCA protein estimation kit (cat. #23227). Finally, all samples were diluted in SDS-PAGE loading buffer (100mM Tris-Cl, 4% (w/v) SDS, 0.02% (w/v) bromophenol blue, 20% (w/v), glycerol and 7.5% (v/v) β-mercaptoethanol) and heated for 5 minutes at 90 °C.

Immunodetection

Protein samples were separated using SDS-PAGE on polyacrylamide gels and transferred to PVDF membranes with a Mini Trans-Blot Cell module (Bio-Rad). The membranes were blocked for 1 hour at room temperature with 5% (w/v) non-fat dry milk in 1X TBST buffer (20mM Tris-Base, pH 7.6, 150mM NaCl, 0.1% (v/v) Tween-20) with gentle orbital shaking. After blocking, the membranes were incubated overnight at 4°C with primary antibodies (1:2000 dilution) in 1X TBST buffer containing 5% (w/v) non-fat dry milk. Following this, the membranes were washed three times for 10 minutes in 1X TBST and then incubated for 1 hour at room temperature with HRP-conjugated secondary antibodies (1:4000 dilution) in 1X TBST with 5% (w/v) non-fat dry milk. After three additional washes in 1X TBST, chemiluminescent substrate was added and bands were detected using the SuperSignal West chemiluminescent substrate (cat. #34579, cat. #34094) and visualized using the Bio-Rad ChemiDoc^{TM} MP imaging system.

Antibodies

HA-Tag (C29F4) Rabbit mAb (cat. #3724), β-actin (8H10D10) Mouse mAb (cat. #3700), Goat anti-Rabbit IgG-HRP conjugate (cat. #7074) and Horse anti-Mouse IgG-HRP conjugate (cat. #7076) were purchased from Cell Signaling Technology. Pgk1 Mouse mAb (cat. #sc-130335) was purchased from Santa Cruz Biotechnology.

Densitometric Analysis

Raw images of Western blots were analysed using ImageJ to normalize target protein expression with the expression of housekeeping proteins including PGK1 or β-actin in order to make densitometric graphs. Statistical analysis was done using unpaired t-test, with error bars representing SEM. All the experiments were performed independently, thrice.

Quantitative Real Time Polymerase Chain Reaction (RT-qPCR) Assay

For *S. cerevisiae*, overnight cultures of *ΔΔpfk1* and *ΔΔadh1* along with wild-type strain ∑1278b were spotted on SLAD and plates were incubated at 30 °C for ~4 days. After 4 days, colonies were isolated and total RNA was extracted using the protocol mentioned in the Ribopure RNA purification kit, yeast (cat. #AM1926). cDNA synthesis was carried out using PrimeScript cDNA synthesis kit (cat. #6110A). Primers used for RT-qPCR were designed using PrimeQuest tool of Integrated DNA Technologies (IDT). RT-qPCR was performed using iTaq Universal SYBR green mastermix (cat. #1725121). Actin was used as housekeeping gene for normalization. The results were analysed using the $2^{-\Delta\Delta CT}$ method (Winer et al., 1999) and statistical analysis was done using one-way ANOVA test, with error bars representing SEM. All the experiments were performed independently, thrice. Primers used in the RT-qPCR experiments are mentioned in Suppl.Table 2.

For *C. albicans*, overnight cultures of *C. albicans* wild-type SC5314 was spotted on SLAD with and without sub-inhibitory concentration of 2DG (0.2% w/v) or *C. albicans ΔΔpfk1* strain along with wild-type SC5314 was spotted on SLAD and plates were incubated at 37 °C for 4 days. After 4 days, colonies were isolated and total RNA was extracted using the protocol mentioned in the Ribopure RNA purification kit, yeast (cat. #AM1926). cDNA synthesis was carried out using PrimeScript cDNA synthesis kit (cat. #6110A). Primers used for RT-qPCR were designed using PrimeQuest tool of Integrated DNA Technologies (IDT). RT-qPCR was performed using iTaq Universal SYBR green mastermix (cat. #1725121). Actin was used as housekeeping gene for normalization. The results were analysed using the $2^{-\Delta\Delta CT}$ method (Winer et al., 1999) and statistical analysis was done using one-way ANOVA test, with error bars representing SEM. All the experiments were performed independently, thrice. Primers used in the RT-qPCR experiments are mentioned in Suppl.Table 2.

In vitro Microbial Survival Assay

In vitro microbial survival assay in macrophages was conducted as described in previous studies with some modifications (Netea et al., 2002; Pradhan et al., 2016). *RAW 264.7* macrophages (1×10^6 cells) were cultured in six-well plates with DMEM supplemented with 10% FBS. After 16 hours of incubation, the cells were treated with *C. albicans* wild-type SC5314 or *ΔΔpfk1* at a multiplicity of infection (MOI) of 10, for 1 hour in antibiotic-free DMEM containing 10% FBS. The media was then removed, and the cells were washed six times with 1X PBS to eliminate any free, non-phagocytosed fungal cells. Next, the cells were lysed with 350 µl of lysis buffer containing 0.025% SDS in 1X PBS. The lysate volume was adjusted to 1 ml with 1X PBS, and a 100 µl undiluted aliquot was plated on YPD agar plate. Serial dilutions of the lysate were also plated on YPD agar plates. These plates were incubated overnight at 37 °C. The following day, colony forming units (CFU) were counted and plotted. The percentage of survival was calculated as the number of CFU in the presence of macrophages divided by the number of CFU in the absence of macrophages. Statistical analysis was done using unpaired t-test, with error bars representing SEM. All the experiments were performed independently, thrice.

In vivo Murine Model of Systemic Candidiasis

In vivo infection experiments were performed as previously described (Majer et al., 2012). Briefly, *C. albicans* wild-type SC5314 and *ΔΔpfk1* were grown overnight in YPD medium at 30 °C. The overnight cultures were back-diluted in 10 ml of YPD liquid medium and allowed to grow till

OD of ~0.6, then washed and resuspended in 1X PBS. C57BL/6 wild-type mice, aged 6 to 8 weeks, were injected intravenously *via* lateral tail vein with *C. albicans* wild-type SC5314 (1×10^7 CFU) or *ΔΔpfk1* (1×10^7 CFU). After infection, mice were monitored for survival and daily for signs of morbidity and mortality were recorded over a period of 21 days. Survival data were analysed using the Log-rank Mantel-Cox test. In order to perform *in vivo* sulfur supplementation, N-acetyl cysteine (NAC) was dissolved in distilled water at a concentration of 6mg/ml and administrated orally to mice before 72 hours of *C. albicans* infection. Normal distilled water administration to mice was used as control. After 72 hours of NAC administration, C57BL/6 wild-type mice, aged 6 to 8 weeks, were infected with SC5314 (1×10^7 CFU) or *ΔΔpfk1* strain (1×10^7 CFU), intravenously *via* lateral tail vein injection. After infection, mice were monitored for survival and daily for signs of morbidity and mortality were recorded over a period of 21 days. Survival data were analysed using the Log-rank Mantel-Cox test.

Determination of Fungal Burden

To assess fungal burden, mice injected with *C. albicans* wild-type SC5314 or *ΔΔpfk1* were sacrificed at ~2 days and kidneys were harvested from infected mice. Kidneys were weighed and then homogenized using a sterilized and 1X PBS-washed Dounce homogenizer (cat. #D8938). Kidney suspensions were then diluted in 1 ml of 1X PBS and serially diluted. Then 100µl of dilutions was plated on YPD agar plates and allowed to grow at 37 °C. Colony forming units (CFU) were calculated by the following formula: (number of colonies × dilution factor × 10)/organ weight in grams (g). Statistical analysis was done using unpaired t-test. Error bars represent SEM. All the experiments were performed independently, thrice.

Histopathology

For histopathological examination, kidneys were collected and fixed in 10% neutral buffered formalin for about two weeks. After fixation, the kidneys were processed, embedded in paraffin, and sectioned to 4µm thickness. The tissue sections were then stained using the GMS stain (Grocott Methenamine Silver stain) as per the manufacturer’s protocol (cat. #AB287884).

Illustrations

Figure illustrations were created using Biorender (*https://app.biorender.com*).

Animal Ethics Statement

All animal experiments in this manuscript were reviewed and approved by the Institutional Animal Ethics Committee (IAEC) of CSIR-Centre for Cellular and Molecular Biology (Approval number: 52/2024-C).

Supporting Information

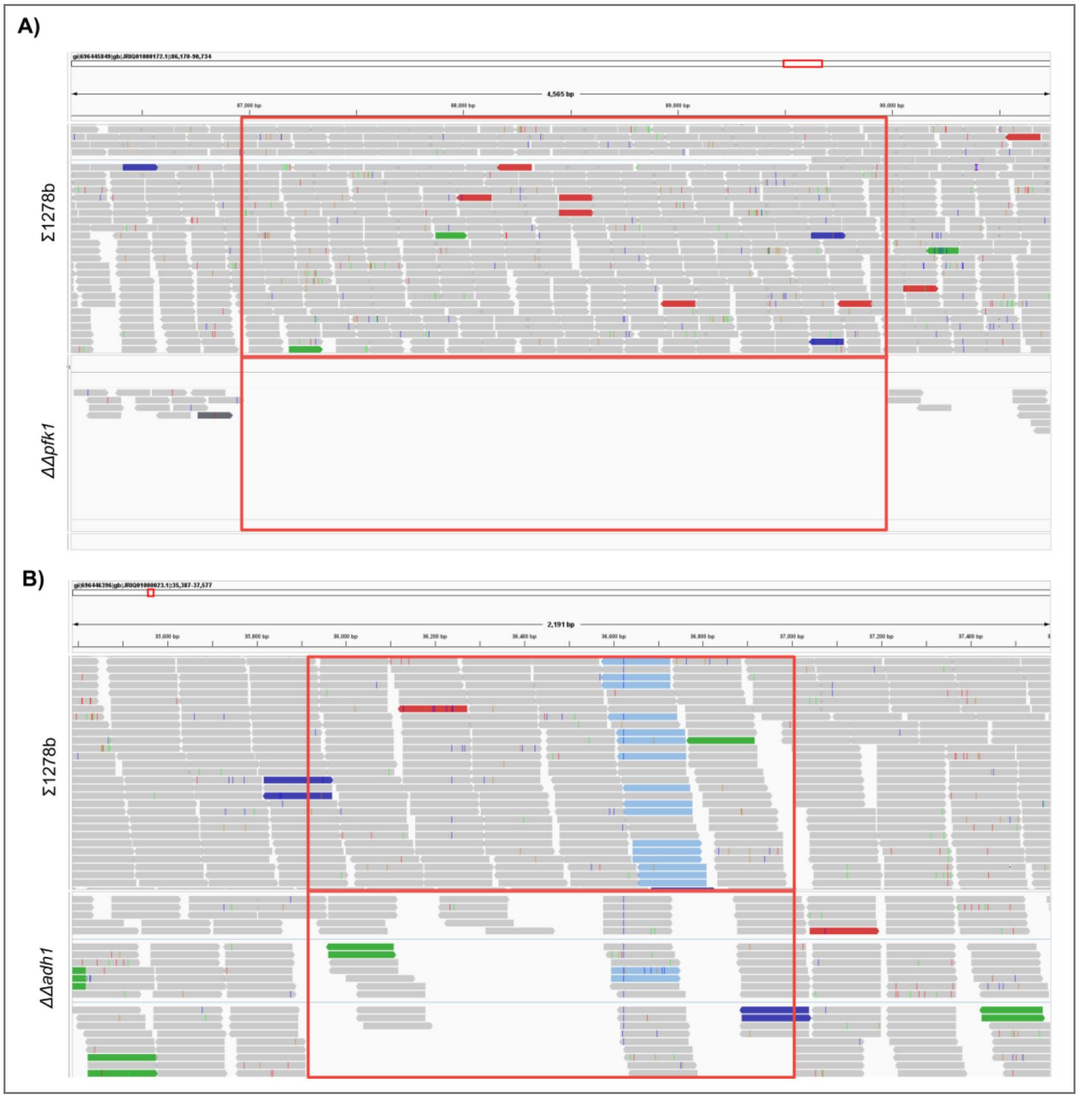


Figure Supplement 1. Confirmation of *pfk1* and *adh1* Deletion in *S. cerevisiae*. **(A)** Confirmation of *ΔΔpfk1* strain using whole genome sequencing. IGV software was used for data visualisation. Red box represents gene locus of *pfk1* in wild-type (∑1278b) and *ΔΔpfk1* strains. **(B)** Confirmation of *ΔΔadh1* strain using whole genome sequencing. IGV software was used for data visualisation. Red box represents gene locus of *adh1* in wild-type (∑1278b) and *ΔΔadh1* strains. This figure was created using *Biorender.com*

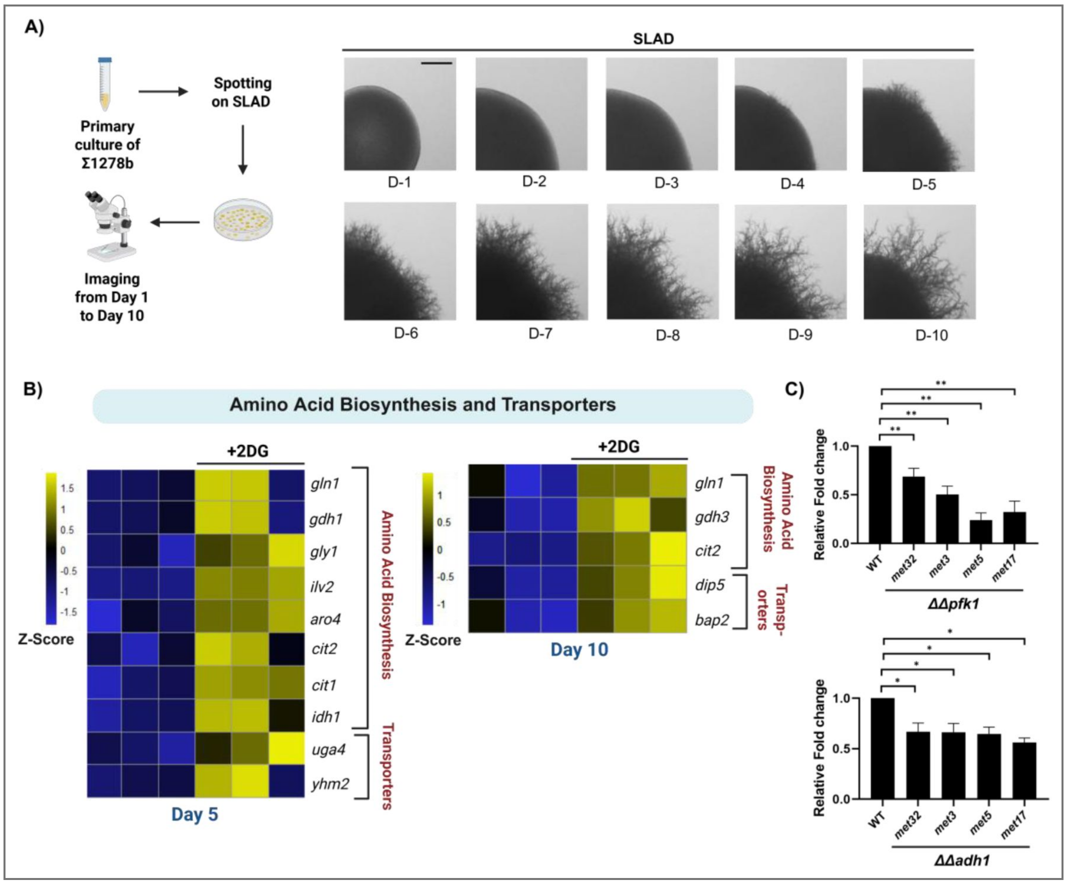


Figure Supplement 2. **Comparative Transcriptomics Identifies the Role of Glycolysis During Pseudohyphal Differentiation in *S. cerevisiae*.**

(A) Wild-type ∑1278b was spotted on SLAD, incubated for 10 days at 30 °C. Colonies were imaged every day, for 10 consecutive days (D-1 to D-10). Scale bar represents 1mm for whole colony images. **(B)** Heatmaps represent the differentially expressed genes involved in amino acid biosynthesis and transporters, in 2DG treated cells compared to untreated cells isolated from D-5 and D-10 colonies, respectively (n=3). Scale bar represents Z-Score. **(C)** *ΔΔpfk1* and *ΔΔadh1* along with Wild-type ∑1278b were spotted on SLAD and cells from colonies were isolated after ∼4 days, for RNA isolation. We then performed comparative RT-qPCR to check for the relative expression of genes involved in the *de novo* biosynthesis of sulfur-containing amino acids pathway including *met32*, *met3*, *met5* and *met17.* Statistical analysis was done using one-way ANOVA test, ** (P<0.01) and *(P<0.05). Error bars represent SEM. This figure was created using *Biorender.com*

Figure Supplement 3. Expression of Proteins Involved in the *de novo* Biosynthesis of Sulfur-containing Amino Acids in Response to Glycolysis Perturbation.

(A) Epitope-tagged strains of various proteins involved in the *de novo* biosynthesis of sulfur-containing amino acids including Met4, Met32, Met16, Met10, Cys4 and Cys3 were spotted on SLAD in the presence and absence of sub-inhibitory concentration of 2DG following which, cells from colonies were isolated at day 5 and levels of these proteins were assessed using Western blotting. Raw images of Western blots were analysed using ImageJ to normalize targeted protein expression with the expression of housekeeping protein- β-actin, in order to generate densitometric graphs. Statistical analysis was done using unpaired t-test, **($P<0.01$) and *($P<0.05$). Error bars represent SEM. **(B)** Epitope-tagged strains of various proteins involved in the *de novo* biosynthesis of sulfur-containing amino acids including Met32, Met16, Met10 and Cys3 were grown in liquid SLAD and then treated with sub-inhibitory concentration of 2DG for 24 hours after which protein expression was checked using Western blotting. Pgk1 was used as loading control. **(C)** Raw images of Western blots were analysed using ImageJ to normalize targeted protein expression with the expression of housekeeping protein-Pgk1, in order to generate densitometric graphs. Statistical analysis was done using unpaired t-test, ***($P<0.001$), **($P<0.01$) and *($P<0.05$). Error bars represent SEM. This figure was created using *Biorender.com*

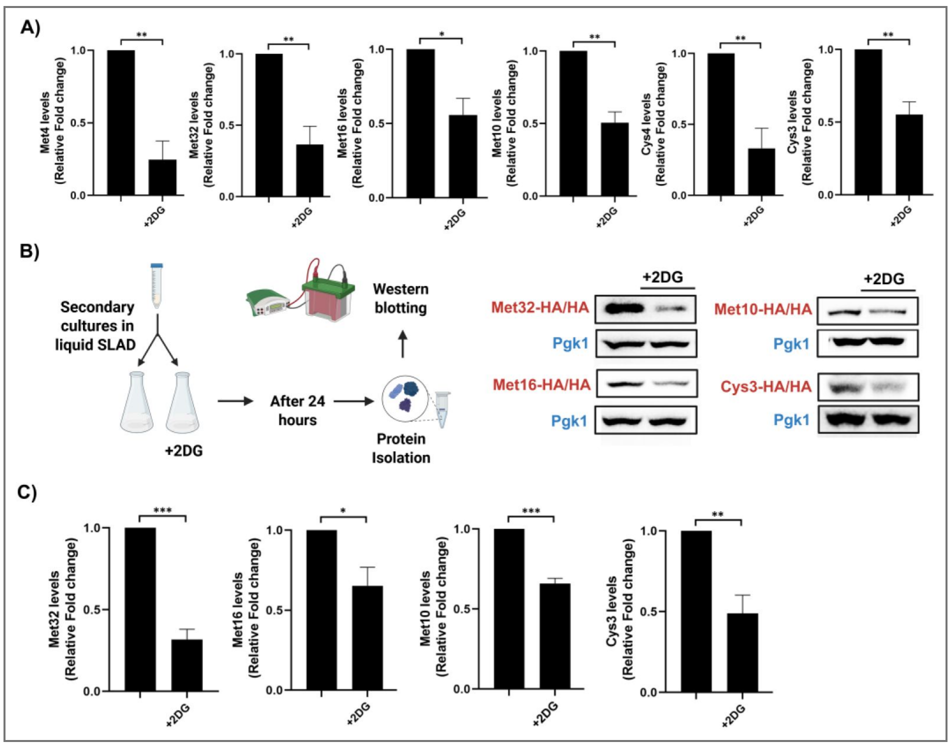
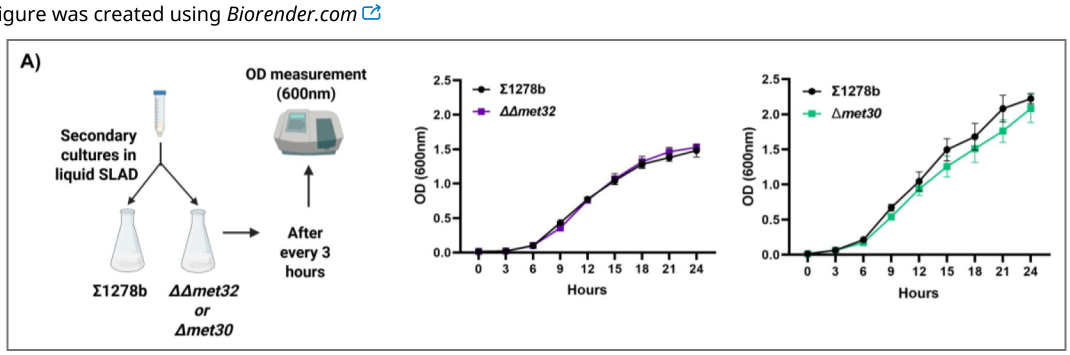


Figure Supplement 4. Monitoring of Overall Growth Rate of *ΔΔmet32* and *Δmet30* Strains.

(A) Growth curve analysis was performed to monitor overall growth of wild-type and *ΔΔmet32* strain as well as wild-type and *Δmet30* strain on SLAD. Overnight grown cultures of deletion strains (*ΔΔmet32* or *Δmet30*) along with wild-type strain (∑1278b) were diluted to OD600=0.01 in fresh SLAD and allowed to grow at 30 °C for 24 hours. OD600 was recorded at 3-h intervals. This figure was created using *Biorender.com*



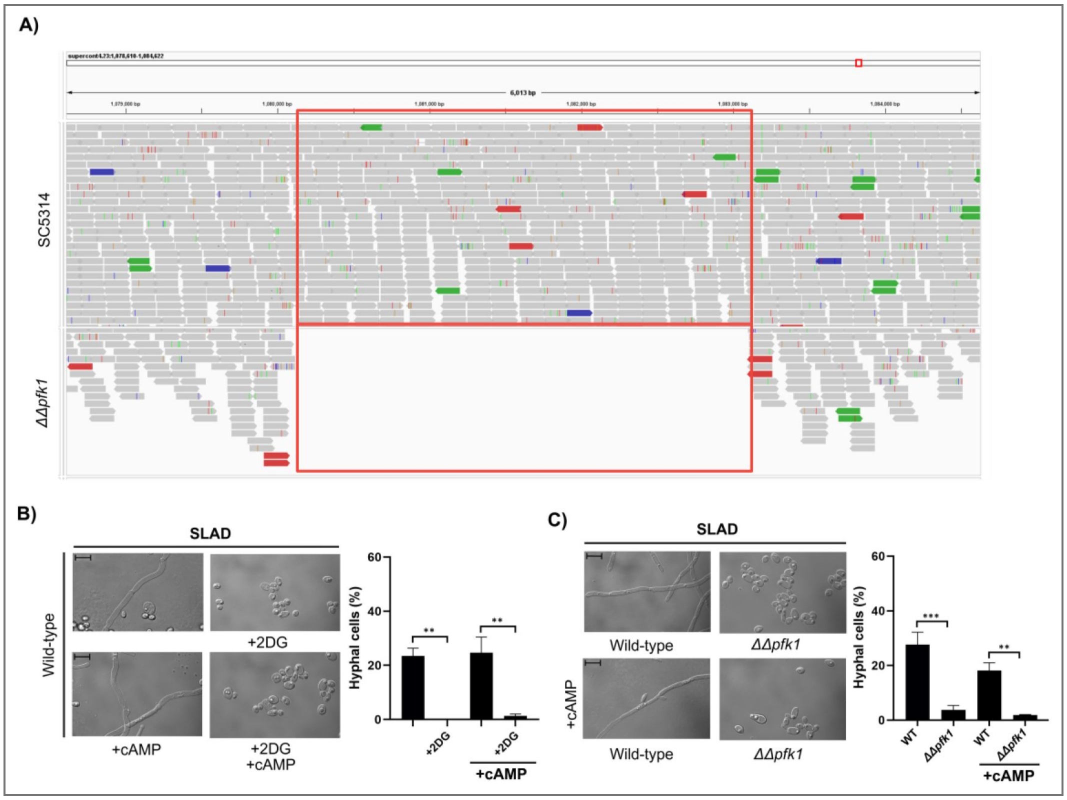


Figure Supplement 5. Confirmation of *pfk1* Deletion in *C. albicans* and Glycolysis is Critical for Hyphal Differentiation in a cAMP-PKA Independent Manner in *C. albicans*.

(A) Confirmation of *ΔΔpfk1* strain using whole genome sequencing. IGV software was used for data visualisation. Red box represents gene locus of *pfk1* in wild-type (SC5314) and *ΔΔpfk1* strains. **(B)** Wild-type SC5314 was spotted on SLAD containing sub-inhibitory concentration of 2DG with and without cAMP (5mM), incubated for 7 days at 37 °C. Cells from colonies were isolated and length/width ratio of individual cells was measured using ImageJ and percentage of hyphal cells from total population was determined. More than 500 cells were counted for each condition. Scale bar represents 10µm for single cell images. Statistical analysis was done using one-way ANOVA test, **(P<0.01). Error bars represent SEM. **(C)** Pertinent knockout strain which lacks the gene that encodes for the glycolytic enzyme phosphofructokinase-1 (*pfk1)* was spotted on SLAD with and without cAMP (5mM), along with wild-type and incubated for 7 days at 37 °C. Cells from colonies were isolated and length/width ratio of individual cells was measured using ImageJ and percentage of hyphal cells from total population was determined. More than 500 cells were counted for each condition. Scale bar represents 10µm for single cell images. Statistical analysis was done using one-way ANOVA test, ***(P<0.001) and **(P<0.01). Error bars represent SEM. This figure was created using *Biorender.com*

Strain	**Genotype**
***S. cerevisiae* Σ1278b**	
Wild-type	*MATa/α*
SV18	*Δadh1::G418/Δadh1::HygB MATa/α*
SV21	*Δpfk1::HygB/Δpfk1::G418 MATa/α*
SV270	*Δmet32::NAT/Δmet32::G418 MATa/α*
SV320	*MET32-HA::G418/MET32-HA::NAT MATa/α*
SV394	*CYS3-HA::NAT/CYS3-HA::G418 MATa/α*
SV395	*CYS4-HA::NAT/CYS4-HA::G418 MATa/α*
SV402	*MET10-HA::NAT/MET10-HA::G418 MATa/α*
SV404	*MET16-HA::NAT/MET16-HA::G418 MATa/α*
SV451	*MET30-HA::NAT/MET30-HA::G418 MATa/α*
SV461	*MET4-HA::NAT/MET4-HA::G418 MATa/α*
SV463	*Δmet30::HygB/MET4-HA::NAT/MET4-HA::G418 MATa/α*
SV473	*MET16-HA::NAT/MET16-HA::G418/Δmet30::HygB MATa/α*
SV475	*MET32-HA::G418/MET32-HA::NAT/Δmet30::HygB MATa/α*
SV476	*CYS3-HA::NAT/CYS3-HA::G418/Δmet30::HygB MATa/α*
***S. cerevisiae* CEN.PK**	
Wild-type	*MATa/α*
SV109	*Δadh1::G418/Δadh1::NAT MATa/α*
SV117	*Δpfk1::NAT/Δpfk1::G418 MATa/α*
***C. albicans* SC5314**	
Wild-type	*MATa/α*
SV308	*Δpfk1::FRT/Δpfk1::SAT-FLP MATa/α*

Supplemental Table 1. Yeast Strains Used in this Study

Oligonucleotides	Sequence (5′ to 3′)
Primers for *S. cerevisiae* Gene Editing and Genotyping	
ADH1 KO F1	AGCTATACCAAGCATACAATCAACTATCTCATATACAATGCGGATCCC CGGGTTAATTAA
ADH1 R1	ATTTAATAATAAAAATCATAAATCATAAGAAATTCGCTTAGAATTCGA GCTCGTTTAAAC
ADH1 KO Chk	CGTTGTTGTCTCACCATATC
PFK1 KO F1	TTTTATATAAAAAATCTGAAACAAAATCATATCAAAGATGCGGATCCC CGGGTTAATTAA
PFK1 R1	CTCCTTTTGCTTAACTTAAACTTTTCATTGCAATCATTCAGAATTCGAG CTCGTTTAAAC

Supplemental Table 2. **Oligonucleotides Used in this Study**

PFK1 KO Chk	AGGATGAGAAAGTGAAATCG
MET32 KO F1	AAGTCAAGAGGTATTATAAATTTCAAAAAAGTACCAAATGCGGATCC CCGGGTTAATTAA
MET32 R1	ATTTTCTTATTTGAGAAGATACACGCTATTTACTCTTTCAGAATTCGA GCTCGTTTAAAC
MET32 KO Chk	GATGGTTTTTCGTCCCTTAC
MET32 C tag F1	TCATCGTCAAGATAACAACCACAATGGTAGCAGTAATGGCCGGATCC CCGGGTTAATTAA
MET32 C tag Chk	GATGGTTTTTCGTCCCTTAC
CYS3 C tag F1	TTTGGAAGACATCAAGCAAGCCTTGAAACAAGCCACCAACCGGATCC CCGGGTTAATTAA
CYS3 R1	AAAGGTCCGGTCGAAGGCAGAGACGTGGCACTGGCGATTAGAATTC GAGCTCGTTTAAAC
CYS3 C tag Chk	CAGAGGTTTGAAGACTTTGC
MET10 Ctag F1	AGAATTAAAGGAAGCATCAAGATACATTTTAGAAGTCTACCGGATCC CCGGGTTAATTAA
MET10 R1	ACTGCTTTTCCTCGAGGTCACCCAAATATACAACGAGATGCGGATCC CCGGGTTAATTAA
MET10 Ctag Chk	TGGTTTAGGTACTGGTTTGG
MET16 Ctag F1	TGAAGCCAGCCGATTCGCGCAATTTTTAAAGCAAGATGCCCGGATCC CCGGGTTAATTAA
MET16 R1	GTCACATACATATGGTTATATATCGTACTCTATCTATCTAGAATTCGA GCTCGTTTAAAC
MET16 Ctag Chk	AATACGGGGATTTCTTATGG
MET4 Ctag F1	AAGCTTAAAGAAGCAAATTTTTGAGAAGGTTCAGAAAGAACGGATCC CCGGGTTAATTAA
MET4 R1	TATATATATATATAATTAAACTGTATAGTCTGTTATTTTAGAATTCGAG CTCGTTTAAAC
MET4 Ctag Chk	GAGCCAACTTTAAATGCAAG
CYS4 Ctag F1	CCATATCGTTACTAAGATGGATTTACTGAGCTACTTAGCACGGATCC CCGGGTTAATTAA
CYS4 R1	TTCTATGTTTGCTTTTATTTGAAGCGTGGGTTCTTATTTAGAATTCGA GCTCGTTTAAAC
CYS4 Ctag Chk	TTAAGGAAACCGCTAAGGTC
MET30 KO F1	AGGGGTGTGTGTTTGGTGATTTATAAAGGAGAAGGGCATGCGGATC CCCGGGTTAATTAA
MET30 R1	CGGATGTTTTTGACCAAGAAAAGACCACACACAGGTCCTAGAATTCG AGCTCGTTTAAAC
MET30 KO Chk	AGAGATAACTGCAGGGTGTG
MET30 Ctag F1	ATTTGGGTGCGTAAAAATGTACAAATTCGATCTCAATGATCGGATCC CCGGGTTAATTAA
MET30 Ctag Chk	AGGATGACCCAACAATGAC
Primers for *C. albicans* Gene Editing and Genotyping	
Psfs2a PFK1 F1	ATTTGAATAGCTTTTTCTTATTATTGAATTCTTTAATTTATCGTTTATTG CAAACAATCAATTATTCCTTATTATTTTGGGCAGATAACGAATTAGAA ATGAAGTTCCTATACTTTCTAG

Supplemental Table 2. (continued)

Psfs2a PFK1 R1	CAACACATTGAATTACATATATGAGTATAAAAAATCAAACTATACTGG GGACTACAAATCTTACTATTTTACACAACTCCTTATCTTCTGTTCTTCA TTTGAAGTTCCTATTCTCTAGAA
Psfs2a SAT-FLP F1	AACATTGGATGCTGAGAACC
Psfs2a SAT-FLP R1	GCACATAATGCTATTTTCTC
PFK1 ApaI UF1	GAGAGGGCCCGTGATGCCTAACTTCTTGGT
PFK1 SacII DR1	CTCTCCGCGGTAGGCTGAATTGCTCGTATG
PFK1 KO R1 Chk	GTACCAATTGAAGTACCACC
FRT F1	CTTTCTAGAGAATAGGAACT
FRT R1	AGTTCCTATTCTCTAGAAAG
Primers for RT-qPCR	
S. cerevisiae ACT1 F1	CGTCTGGATTGGTGGTTCTATC
S. cerevisiae ACT1 R1	GGACCACTTTCGTCGTATTCTT
S. cerevisiae MET32 F1	TCCTCAATGTGGCAAAGGT
S. cerevisiae MET32 R1	CACCACCCGCAGTTAGTAAT
S. cerevisiae MET3 F1	CAGAGTTGAGACGCCGTTTA
S. cerevisiae MET3 R1	GTCTTGGTGGGTTGGATTCT
S. cerevisiae MET5 F1	CATCGCCGTTCCTCCATATAAC
S. cerevisiae MET5 R1	CCCATACCACCACCAACAAA
S. cerevisiae MET17 F1	CCCTGGTTTAGCATCTCATTCT
S. cerevisiae MET17 R1	CAGTTTCCTTGTCGGCATTTG
C. albicans ACT1 F1	TTGGATTCTGGTGATGGTGTTA
C. albicans ACT1 R1	TCAAGTCTCTACCAGCCAAATC
C. albicans MET32 F1	CACCAACACAACACCATCAAC
C. albicans MET32 R1	TCCAATTCCACCACCAGTAATC
C. albicans MET3 F1	GTTCCAACAACAAACCCAACTC
C. albicans MET3 R1	TCACCATCAGCATTACCAGAAG
C. albicans MET5 F1	TTTATGGTGCGTGTTCGTTTAC
C. albicans MET5 R1	TAGCCCTGGTGGTCAATTTC
C. albicans MET10 F1	GGGTTGATCCAAAGGGAAGAT
C. albicans MET10 R1	GACAATGGTGGCAACTTCATAAC
C. albicans MET17 F1	TCAAGGACATCACCAACACC
C. albicans MET17 R1	CTTAGAGTCACCCACATTAGCC
C. albicans ALS3 F1	GGTGGCACTGATTCGGTTAT
C. albicans ALS3 R1	GCGGTAATGGTAGTGGTAGTTG
C. albicans ECE1 F1	TTGGCAACATTCCACAAGTAATC
C. albicans ECE1 R1	CAGCAATGATACCAGCAACAAC
C. albicans HWP1 F1	CTCTACGACTGAAGGTGCTATTC
C. albicans HWP1 R1	GTAGAAATAGGAGCGACACTTGA
C. albicans HYR1 F1	CACATCAAGTCCTGGTCAATCTA
C. albicans HYR1 R1	TGGAACAGTGGTGAAGATAGTG
C. albicans IHD1 F1	GGCTCTCAAGGTCAATCTACAA

Supplemental Table 2. (continued)

C. albicans IHD1 R1	ACTAGCACCATCGTTACCATTAG
C. albicans RBT1 F1	CCATCTGCTAACTCCTCATACAC
C. albicans RBT1 R1	CAAGAATGCAGCAAGACCAATAA
C. albicans SAP6 F1	GAGTGTTCTTGCTTTCGCTTTAT
C. albicans SAP6 R1	CATCTGGATCAACAAGGGATCT

Supplemental Table 2. (continued)

Reagents/ Resource	Reference or Source	Identifier or catalog Number
Recombinant DNA		
pFA6a-kanMX	Gift from Dr. Sunil Laxman	Not applicable
pFA6a-Hyg	Gift from Dr. Sunil Laxman	Not applicable
pFA6a-Nat	Gift from Dr. Sunil Laxman	Not applicable
pFA6a-Ctag-HA-G418	Gift from Dr. Sunil Laxman	Not applicable
pFA6a-Ctag-HA-NAT	Gift from Dr. Sunil Laxman	Not applicable
pSFS2a-NAT	Gift from Dr. Kaustuv Sanyal	Not applicable
Antibodies		
Rabbit anti-HA	Cell Signaling Technology	3724S
Mouse anti-β-actin	Cell Signaling Technology	3700
Mouse anti-PGK1	Santa Cruz Biotechnology	sc-130335
Goat anti-Rabbit IgG-HRP conjugate	Cell Signaling Technology	7074S
Horse anti-Mouse IgG-HRP conjugate	Cell Signaling Technology	7076S
Chemicals, enzymes and other reagents		
BD Bacto agar	Becton Dickinson	214010
Agarose	MP Biomedicals	100267
Ammonium sulfate	Sisco Research Laboratories	0149175
Blotto, non-fat dry milk	Santa Cruz Biotechnology	sc-2324
B-mercaptoethanol	Sigma-Aldrich	444203
Diluent	HiMedia	MB228
D-(+)-Glucose anhydrous	HiMedia	GRM016
Lithium acetate	Sigma-Aldrich	517992
Peptone	Thermo Fisher Scientific	211677
RNaseZAP	Sigma-Aldrich	R2020
SDS	G Biosciences	RC1184
Trichloro acetic acid	Sigma-Aldrich	100807
Tris, free base	HiMedia	MB029
Tween 20	Sigma-Aldrich	P7949
Yeast extract	Thermo Fisher Scientific	212750
Yeast nitrogen base w/o amino acids and ammonium sulfate	Becton Dickinson	233520
L-Cysteine	Sigma-Aldrich	30089
L-Methionine	Sigma-Aldrich	64319
Sodium citrate	Sigma-Aldrich	C8532

Supplemental Table 3. **List of Reagents and Tools Used in this Study**

Glycerol	HiMedia	MB060
Potassium acetate	MP Biomedicals	191425
2-Deoxy-D-Glucose	Sigma-Aldrich	D8375
NAT	Jena Bioscience	AB-102XL
G418	MP Biomedicals	158782
Ethidium bromide	HiMedia	MB074
Pico-ECL substrate	Thermo Fisher Scientific	34579
Femto-ECL substrate	Thermo Fisher Scientific	34094
TEMED	Sisco Research Laboratories	52145
Hygromycin	Sigma-Aldrich	H0654
Ribopure RNA purification kit-Yeast	Thermo Fisher Scientific	AM1926
BCA Protein Assay kit	Thermo Fisher Scientific	23227
GMS Staining kit	ABcam	AB287884
Softwares		
RStudio v4.3.2	Posit	Not applicable
GraphPad Prism v9	GraphPad software	Not applicable
ImageJ		Not applicable
IGV v2.16.2		Not applicable
Other		
Glass beads	HiMedia	GRM2228
MATRIX thermoshaker	IKA	Not applicable
Spectrophotometer	Thermo Fisher Scientific	Not applicable
Proflex PCR systems	Thermo Fisher Scientific	Not applicable
MVX10 Stereo Microscope	OLYMPUS Scientific	Not applicable
Biosafety Cabinet	IGene Labserve	Not applicable
PVDF membrane	Milipore	IPVH00010
Mini Trans-Blot Cell Module	Bio-Rad	Not applicable
ChemiDoc™ MP analyzer	Bio-Rad	Not applicable
QuantStudio 5 Real-Time PCR System	Thermo Fisher Scientific	Not applicable
Apotome widefield fluorescence microscope	Zeiss	Not applicable
Centrifuge 5425R	Eppendorf	Not applicable

Supplemental Table 3. (continued)

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Additional information

Data availability

All relevant data are within the paper and its supporting information files, except for RNA-Seq data. RNA-Seq data are deposited in NCBI's SRA database and are accessible through BioProject accession number PRJNA1263201. (*https://www.ncbi.nlm.nih.gov/bioproject/PRJNA1263201*)

Author Contributions

Dhrumi Shah: Conceptualization; Investigation; Formal analysis; Visualization; Writing-original draft. **Nikita Rewatkar:** Investigation. **Adishree M:** Investigation. **Siddhi Gupta:** Investigation. **Sudharsan Mathivathanan:** Investigation. **Sayantani Biswas:** Investigation. **Sriram Varahan:** Conceptualization; Supervision; Funding acquisition; Investigation; Formal analysis; Visualization; Writing-original draft; Writing-review and editing.

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Council of Scientific and Industrial Research, India (CSIR)	FTT070505	Sriram Varahan
Wellcome Trust DBT India Alliance (India Alliance)	IA/E/16/1/502996	Sriram Varahan

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Peer reviews

Reviewer #1 (Public review):

Summary:

Fungal survival and pathogenicity rely on the ability to undergo reversible morphological transitions, which is often linked to nutrient availability. In this study, the authors uncover a conserved connection between glycolytic activity and sulfur amino acid biosynthesis that drives morphogenesis in two fungal model systems. By disentangling this process from canonical cAMP signaling, the authors identify a new metabolic axis that integrates central carbon metabolism with developmental plasticity and virulence.

Strengths:

The study integrates different experimental approaches, including genetic, biochemical, transcriptomic and morphological analyses and convincingly demonstrates that perturbations in glycolysis alters sulfur metabolic pathways and thus impacts pseudohyphal and hyphal differentiation. Overall, this work offers new and important insights into how metabolic fluxes are intertwined with fungal developmental programs and therefore opens new perspectives to investigate morphological transitioning in fungi.

Importantly, in the revised version the authors now substantiate the transcriptomic findings by RT-qPCR analyses in the pfk1ΔΔ and adh1ΔΔ strains, demonstrating that genetic disruption of glycolytic flux generally mirrors the effects of 2-deoxyglucose treatment. The manuscript's discussion has also been strengthened by explicitly addressing why cysteine and methionine differ in their ability to rescue filamentation in S. cerevisiae versus C. albicans, highlighting species-specific differences in sulfur uptake and transsulfuration pathways.

Overall, this revised manuscript provides compelling evidence for a previously unrecognized coupling between glycolysis and sulfur metabolism that shapes fungal morphogenesis and virulence. It opens new perspectives on metabolic control of fungal development and raises interesting mechanistic questions for future work.

Comments on revisions:

The authors have incorporated all of my suggested changes and addressed all raised concerns.

https://doi.org/10.7554/eLife.109075.2.sa3

Reviewer #2 (Public review):

Summary:

The manuscript investigates the interplay between glycolysis and sulfur metabolism in regulating fungal morphogenesis and virulence. Using both Saccharomyces cerevisiae and Candida albicans, the authors demonstrate that glycolytic flux is essential for morphogenesis under nitrogen-limiting conditions, acting independently of the established cAMP-PKA pathway. Transcriptomic and genetic analyses reveal that glycolysis influences the de novo biosynthesis of sulfur-containing amino acids, specifically cysteine and methionine. Notably, supplementation with sulfur sources restores morphogenetic and virulence defects in glycolysis-deficient mutants, thereby linking core carbon metabolism with sulfur assimilation and fungal pathogenicity.

Strengths:

The work identifies a previously uncharacterized link between glycolysis and sulfur metabolism in fungi, bridging metabolic and morphogenetic regulation which is an important conceptual advance and fungal pathogenicity. Demonstrating that adding cysteine supplementation rescues virulence defects in animal model connects basic metabolism to infection outcomes that add on biomedical importance.

Comments on revisions:

The authors have sufficiently addressed my concern and provided a clear justification for their proposed model including the limitations of performing the mechanistic assays at this stage. I am satisfied with the response and have no further comments

https://doi.org/10.7554/eLife.109075.2.sa2

Reviewer #3 (Public review):

This study investigates the connection between glycolysis and the biosynthesis of sulfur-containing amino acids in controlling fungal morphogenesis, using Saccharomyces cerevisiae and C. albicans as model organisms. The authors identify a conserved metabolic axis that integrates glycolysis with cysteine/methionine biosynthetic pathways to influence morphological transitions. This work broadens the current understanding of fungal morphogenesis, which has largely focused on gene regulatory networks and cAMP-dependent signaling pathways, by emphasizing the contribution of metabolic control mechanisms.

Strengths:

The delineation of how glycolytic flux regulates fungal morphogenesis through a cAMP-independent mechanism is an advancement. The coupling of glycolysis with the de novo biosynthesis of sulfur-containing amino acids, a requirement for morphogenesis, introduces a novel and unexpected layer of regulation.

Demonstrating this mechanism in both S. cerevisiae and C. albicans strengthens the argument for its evolutionary conservation and biological importance.

The ability to rescue the morphogenesis defect through supplementation of sulfur-containing amino acids provides a functional validation.

Weaknesses:

cAMP addition rescued the pseudohyphal differentiation defect exhibited by the ΔΔgpa2 strain. More clarity is needed on how this mechanism is mechanistically distinct from the metabolic control - whether cAMP acts in parallel or downstream to sulfur-containing amino acids biosynthesis has to be characterized. Supplementation of cysteine and methionine bypasses glycolytic regulation; the link between these amino acids and their role in fungal morphogenesis is not completely characterized.

The demonstrated link between glycolysis and sulfur amino acid biosynthesis, along with its implications for virulence in C. albicans, is important for understanding fungal adaptation, as mentioned in the article; however, the downstream effects of Met4 activation were not fully characterized. How does Cysteine/Methionine rescue morphogenesis? The author's response figure 1 shows that there are no significant transcriptional changes in the expression of cAMP-PKA pathway-associated genes, which alone could not completely explain the role of gpa2 in morphogenesis, because exogenous cAMP can restore pseudohyphal differentiation in the ΔΔgpa2 background (Revised Fig. 1L). This implies that gpa2's function in morphogenesis is an additional, or possibly a metabolic or post-transcriptional, layer of regulation, and its connection to sulfur-containing amino acids remains to be elucidated.

https://doi.org/10.7554/eLife.109075.2.sa1

Author response:

The following is the authors' response to the original reviews.

> ***Public Reviews:***
>
> ***Reviewer #1 (Public review):***
>
> *Summary:*
>
> *Fungal survival and pathogenicity rely on the ability to undergo reversible morphological transitions, which are often linked to nutrient availability. In this study, the authors uncover a conserved connection between glycolytic activity and sulfur amino acid biosynthesis that drives morphogenesis in two fungal model systems. By disentangling this process from canonical cAMP signaling, the authors identify a new metabolic axis that integrates central carbon metabolism with developmental plasticity and virulence.*
>
> *Strengths:*
>
> *The study integrates different experimental approaches, including genetic, biochemical, transcriptomic, and morphological analyses, and convincingly demonstrates that perturbations in glycolysis alter sulfur metabolic pathways and thus impact pseudohyphal and hyphal differentiation. Overall, this work offers new and important insights into how metabolic fluxes are intertwined with fungal developmental programs and therefore opens new perspectives to investigate morphological transitioning in fungi.*

We thank the reviewer for finding this study to be of importance and for appreciating our multipronged approach to substantiate our finding that perturbations in glycolysis alter sulfur metabolism and thus impact pseudohyphal and hyphal differentiation in fungi.

> *Weaknesses:*
>
> *A few aspects could be improved to strengthen the conclusions. Firstly, the striking transcriptomic changes observed upon 2DG treatment should be analyzed in S. cerevisiae adh1 and pfk1 deletion strains, for instance, through qPCR or western blot analyses of sulfur metabolism genes, to confirm that observed changes in 2DG conditions mirror those seen in genetic mutants. Secondly, differences between methionine and cysteine in their ability to rescue the mutant phenotype in both species are not mentioned, nor discussed in more detail. This is especially important as there seem to be differences between S. cerevisiae and C. albicans, which might point to subtle but specific metabolic adaptations.*
>
> *The authors are also encouraged to refine several figure elements for clarity and comparability (e.g., harmonized axes in bar plots), condense the discussion to emphasize the conceptual advances over a summary of the results, and shorten figure legends.*

We are grateful for this valuable and constructive feedback, and we agree with the reviewer on the necessity of performing RT-qPCR analysis of sulfur metabolism genes in ΔΔpfk1 and

ΔΔadh1 strains of S. cerevisiae to validate our RNA-Seq results using 2DG. We have performed this experiment, and our results show that several genes involved in the de novo biosynthesis of sulfur-containing amino acids are downregulated in both the ΔΔpfk1 and ΔΔadh1 strains, corroborating the downregulation of sulfur metabolism genes in the 2DG treated samples. This new data is now included in the revised manuscript as Supplementary Figure 2C.

Furthermore, we acknowledge the reviewer's point regarding the significance of comparing the differences in the ability of methionine and cysteine to rescue filamentation defects exhibited by the mutants, between S. cerevisiae and C. albicans. The observed differences between S. cerevisiae and C. albicans likely highlight species-specific metabolic adaptations within the sulfur assimilation pathway. While both yeasts employ the transsulfuration pathway to interconvert these sulfur-containing amino acids, the precise regulatory points including the specific enzymes, their compartmentalization, and transcriptional control are not identical. For instance, differences in the feedback inhibition mechanisms or the expression levels of key transsulfuration enzymes between S. cerevisiae and C. albicans could explain the variations in the phenotypic rescue experiments (Chebaro et al., 2017; Lombardi et al., 2024; Rouillon et al., 2000; Shrivastava et al., 2021; Thomas and Surdin-Kerjan, 1997). Furthermore, the species-specific differences in amino acid transport systems (permeases) adds another layer of complexity. S. cerevisiae primarily uses multiple, low-affinity permeases for cysteine transport (Gap1, Bap2, Bap3, Tat1, Tat2, Agp1, Gnp1, Yct1), while relying on a limited set of high-affinity transporters (like Mup1) for methionine transport, with the added complexity that its methionine transporters can also transport cysteine (Düring-Olsen et al., 1999; Huang et al., 2017; Kosugi et al., 2001; Menant et al., 2006). In contrast, C. albicans utilizes a high-affinity transporters for the uptake of both amino acids, employing Cyn1 specifically for cysteine and Mup1 for methionine, indicating a greater reliance on dedicated transport mechanisms for these sulfur-containing molecules in the pathogenic yeast (Schrevens et al., 2018; Yadav and Bachhawat, 2011). A combination of the aforesaid factors could be the potential reason for the differences in the ability of cysteine and methionine to rescue filamentation in S. cerevisiae and C. albicans.

Finally, we have enhanced the quantitative rigor and clarity of the data presentation in the revised manuscript by implementing Y-axis uniformity across all relevant bar graphs to facilitate a more robust and direct comparative analysis. We have also condensed the discussion to emphasize the conceptual advances and have shortened the figure legends as per the reviewer suggestions

> ***Reviewer #2 (Public review):***
>
> *Summary:*
>
> *This manuscript investigates the interplay between glycolysis and sulfur metabolism in regulating fungal morphogenesis and virulence. Using both Saccharomyces cerevisiae and Candida albicans, the authors demonstrate that glycolytic flux is essential for morphogenesis under nitrogen-limiting conditions, acting independently of the established cAMP-PKA pathway. Transcriptomic and genetic analyses reveal that glycolysis influences the de novo biosynthesis of sulfur-containing amino acids, specifically cysteine and methionine. Notably, supplementation with sulfur sources restores morphogenetic and virulence defects in glycolysis-deficient mutants, thereby linking core carbon metabolism with sulfur assimilation and fungal pathogenicity.*
>
> *Strengths:*
>
> *The work identifies a previously uncharacterized link between glycolysis and sulfur metabolism in fungi, bridging metabolic and morphogenetic regulation, which is an important conceptual advance and fungal pathogenicity. Demonstrating that adding*

> *cysteine supplementation rescues virulence defects in animal models connects basic metabolism to infection outcomes, which adds to biomedical importance.*

We would like to thank the reviewer for the positive comments on our work. We are pleased that they recognize the novel metabolic link between glycolysis and sulfur metabolism as a key conceptual advance in fungal morphogenesis.

> *Weaknesses:*
>
> *The proposed model that glycolytic flux modulates Met30 activity post-translationally remains speculative. While data support Met4 stabilization in met30 deletion strains, the mechanism of Met30 modulation by glycolysis is not demonstrated.*

We thank the reviewer for this valuable feedback. The activity of the SCF^{Met30} E3 ubiquitin ligase, mediated by the F box protein Met30, is dynamically regulated through both proteolytic degradation and its dissociation from the SCF complex, to coordinate sulfur metabolism and cell cycle progression (Smothers et al., 2000; Yen et al., 2005). Our transcriptomic (RNA-seq analysis) and protein expression analysis (Fig. 3J) confirms that Met30 expression is not differentially regulated in the presence of 2DG, effectively eliminating changes in synthesis or SCF^{Met30} proteasomal degradation as the dominant regulatory mechanism. This observation is consistent with the established paradigm wherein stress signals, such as cadmium (Cd^{2+}) exposure, rapidly inactivates the SCF^{Met30} E3 ubiquitin ligase via the dissociation of Met30 from the Skp1 subunit of the SCF complex (Lauinger et al., 2024; Yen et al., 2005). We therefore propose that active glycolytic flux modulates SCF^{Met30} activity post-translationally, specifically by triggering Met30 detachment from the SCF complex. This mechanism would stabilize the primary substrate, the transcription factor Met4, thus promoting the biosynthesis of sulfur-containing amino acids. Mechanistic validation of this hypothesis, particularly the assessment of Met30 dissociation from the SCF^{Met30} complex via immunoprecipitation (IP), is technically challenging. Since these experiments will involve isolation of cells from colonies undergoing pseudohyphal differentiation, on solid media (given that pseudohyphal differentiation does not occur in liquid media that is limiting for nitrogen (Gancedo, 2001; Gimeno et al., 1992)), current cell yields (OD_{600}≈1 from ≈80-100 colonies) are significantly below the amount of cells that is needed to obtain the required amount of total protein concentration, for standard pull down assays (OD_{600}≈600-800 is required to achieve 1-2 mg/ml of total protein which is the standard requirement for pull down protocols in S. cerevisiae (Lauinger et al., 2024)).

Given that the primary objective of our study is to establish the novel regulatory link between glycolysis and sulfur metabolism in the context of fungal morphogenesis, we would like to explore these crucial mechanistic details, in depth, in a subsequent study.

> ***Reviewer #3 (Public review):***
>
> *This study investigates the connection between glycolysis and the biosynthesis of sulfur-containing amino acids in controlling fungal morphogenesis, using Saccharomyces cerevisiae and C. albicans as model organisms. The authors identify a conserved metabolic axis that integrates glycolysis with cysteine/methionine biosynthetic pathways to influence morphological transitions. This work broadens the current understanding of fungal morphogenesis, which has largely focused on gene regulatory networks and cAMP-dependent signaling pathways, by emphasizing the contribution of metabolic control mechanisms. However, despite the novel conceptual framework, the study provides limited mechanistic characterization of how the sulfur metabolism and glycolysis blockade directly drive morphological outcomes. In particular, the rationale for selecting specific gene deletions, such as Met32 (and not Met4), or the Met30 deletion used to probe this pathway, is not clearly explained, making it difficult to assess whether these targets comprehensively represent the metabolic nodes proposed to be critical.*

> *Further supportive data and experimental validation would strengthen the claims on connections between glycolysis, sulfur amino acid metabolism, and virulence.*
>
> *Strengths:*
>
> *(1) The delineation of how glycolytic flux regulates fungal morphogenesis through a cAMP-independent mechanism is a significant advancement. The coupling of glycolysis with the de novo biosynthesis of sulfur-containing amino acids, a requirement for morphogenesis, introduces a novel and unexpected layer of regulation.*
>
> *(2) Demonstrating this mechanism in both S. cerevisiae and C. albicans strengthens the argument for its evolutionary conservation and biological importance.*
>
> *(3) The ability to rescue the morphogenesis defect through exogenous supplementation of sulfur-containing amino acids provides functional validation.*
>
> *(4) The findings from the murine Pfk1-deficient model underscore the clinical significance of metabolic pathways in fungal infections.*

We are grateful for this comprehensive and insightful summary of our work. We deeply appreciate the reviewer's recognition of the key conceptual breakthroughs regarding the metabolic regulation of fungal morphogenesis and the clinical relevance of our findings.

> *Weaknesses:*
>
> *(1) While the link between glycolysis and sulfur amino acid biosynthesis is established via transcriptomic and proteomic analysis, the specific regulation connecting these pathways via Met30 remains to be elucidated. For example, what are the expression and protein levels of Met30 in the initial analysis from Figure 2? How specific is this effect on Met30 in anaerobic versus aerobic glycolysis, especially when the pentose phosphate pathway is involved in the growth of the cells when glycolysis is perturbed ?*

We are grateful for the insightful feedback provided by the reviewer. S. cerevisiae is a Crabtree positive organism that primarily uses anaerobic glycolysis to metabolize glucose, under glucose-replete conditions (Barford and Hall, 1979; De Deken, 1966) and our pseudohyphal differentiation assays are performed in glucose-rich conditions (Gimeno et al., 1992). Furthermore, perturbation of glycolysis is known to induce compensatory upregulation of the Pentose Phosphate Pathway (PPP) (Ralser et al., 2007) and we have also observed the upregulation of the gene that encodes for transketolase-1 (Tkl1), a key enzyme in the PPP, in our RNA-seq data. Importantly, our transcriptomic (RNA-seq analysis) and protein expression analysis (Fig. 3J) confirms that Met30 expression is not differentially regulated in the presence of 2DG, effectively eliminating changes in synthesis or SCF^{Met30} proteasomal degradation as the dominant regulatory mechanism. This aligns with the established paradigm wherein stress signals, such as cadmium (Cd^{2+}) exposure, rapidly inactivates SCF^{Met30} E3 ubiquitin ligase via Met30 dissociation from the Skp1 subunit of the complex (Lauinger et al., 2024; Yen et al., 2005). We therefore propose that active glycolytic flux modulates SCF^{Met30} activity post-translationally, specifically by triggering Met30 detachment from the SCF complex. This mechanism would stabilize the primary substrate, the transcription factor Met4, thus promoting the biosynthesis of sulfur-containing amino acids. Further experiments are required to delineate the specific role of pentose phosphate pathway in the aforesaid proposed regulation of the Met30 activity under glycolysis perturbation and this will be explored in our subsequent study.

> *(2) Including detailed metabolite profiling could have strengthened the metabolic connection and provided additional insights into intermediate flux changes, i.e., measuring levels of metabolites to check if cysteine or methionine levels are influenced intracellularly. Also, it is expected to see how Met30 deletion could affect cell growth.*

> *Data on Met30 deletion and its effect on growth are not included, especially given that a viable heterozygous Met30 strain has been established. Measuring the cysteine or methionine levels using metabolomic analysis would further strengthen the claims in every section.*

We are grateful to the reviewer for this constructive feedback. To address the potential impact of met30 deletion on cell growth, we have included new data (Suppl. Fig. 4A) demonstrating that the deletion of a single copy of met30 in diploid S. cerevisiae does not compromise overall cell growth under nitrogen-limiting conditions as the Δmet30 strain grows similar to the wild-type strain.

Our pseudohyphal/hyphal differentiation assays show that the defects induced by glycolytic perturbation is fully rescued by the exogenous supplementation of sulfur-containing amino acids, cysteine or methionine. Since these data conclusively demonstrate that the primary metabolic limitation caused by the perturbation of glycolysis, which leads to filamentation defects is sulfur metabolism, we posit that performing comprehensive metabolic profiling would primarily reconfirm the aforesaid results. We believe that our in vitro and in vivo sulfur add-back experiments sufficiently substantiate the novel regulatory metabolic link between glycolysis and sulfur metabolism.

> *(3) In comparison with the previous bioRxiv (doi: https://doi.org/10.1101/2025.05.14.654021) of this article in May 2025 to the recent bioRxiv of this article (doi: https://doi.org/10.1101/2025.05.14.654021), there have been some changes, and Met30 deletion has been recently included, and the chemical perturbation of glycolysis has been added as new data. Although the changes incorporated in the recent version of the article improved the illustration of the hypothesis in Figure 6, which connects glycolysis to Sulfur metabolism, the gene expression and protein levels of all genes involved in the illustrated hypothesis are not consistently shown. For example, in some cases, the Met4 expression is not shown (Figure 4), and the Met30 expression is not shown during profiling (gene expression or protein levels) throughout the manuscript. Lack of consistency in profiling the same set of key genes makes understanding more complicated.*

We thank the reviewer for this feedback which helps us to clarify the scope of our transcriptomic analysis. Our decision to focus our RT-qPCR experiments on downstream targets, while excluding met4 and met30 from the RT-qPCR analysis, is based on their known regulatory mechanisms. Met4 activity is predominantly regulated by post-translational ubiquitination by the SCFMet30 complex followed by its degradation (Rouillon et al., 2000; Shrivastava et al., 2021; Smothers et al., 2000) while Met30 activity is primarily regulated by its auto-degradation or its dissociation from the SCFMet30 complex (Lauinger et al., 2024; Smothers et al., 2000; Yen et al., 2005). Consistent with this, our RNA-Seq results indicate that neither met4 nor met30 transcripts are differentially expressed, in response to 2DG addition. For all our RT-qPCR analysis in S. cerevisiae and C. albicans, we have consistently used the same set of sulfur metabolism genes and these include met32, met3, met5, met10 and met17. Our data on protein expression analysis of Met30 in S. cerevisiae (Fig. 3J) confirms that Met30 expression is not differentially regulated in the presence of 2DG, effectively eliminating changes in synthesis or SCFMet30 proteasomal degradation as the dominant regulatory mechanism.

> *(4) The demonstrated link between glycolysis and sulfur amino acid biosynthesis, along with its implications for virulence in C. albicans, is important for understanding fungal adaptation, as mentioned in the article; however, the Met4 activation was not fully characterized, nor were the data presented when virulence was assessed in Figure 4. Why is Met4 not included in Figure 4D and I? Especially, according to Figure 6, Met4 activation is crucial and guides the differences between glycolysis-active and inactive conditions.*

We thank the reviewer for their input. As the Met4 transcription factor in C. albicans is primarily regulated post-translationally through its degradation and inactivation by the SCFMet30 E3 ubiquitin ligase complex (Shrivastava et al., 2021), we opted to monitor the transcriptional status of downstream targets of Met4 (i.e., genes directly regulated by Met4), as these are the genes that exhibit the most direct and functionally relevant transcriptional changes in response to the altered Met4 levels.

> *(5) Similarly, the rationale behind selecting Met32 for characterizing sulfur metabolism is unclear. Deletion of Met32 resulted in a significant reduction in pseudohyphal differentiation; why is this attributed only to Met32? What happens if Met4 is deleted? It is not justified why Met32, rather than Met4, was chosen. Figure 6 clearly hypothesizes that Met4 activation is the key to the mechanism.*

We sincerely thank the reviewer for this insightful query regarding our selection of the met32 for our gene deletion experiments. The choice of ΔΔmet32 strain was strategically motivated by its unique phenotypic properties within the de novo biosynthesis of sulfur-containing amino acids pathway. While deletions of most the genes that encode for proteins involved in the de novo biosynthesis of sulfurcontaining amino acids, result in auxotrophy for methionine or cysteine, ΔΔmet32 strain does not exhibit this phenotype (Blaiseau et al., 1997). This key distinction is attributed to the functional redundancy provided by the paralogous gene, met31 (Blaiseau et al., 1997). Crucially, given that the deletion of the central transcriptional regulator, met4, results in cysteine/methionine auxotrophy, the use of the ΔΔmet32 strain provides an essential, viable experimental model for investigating the role of sulfur metabolism during pseudohyphal differentiation in S. cerevisiae.

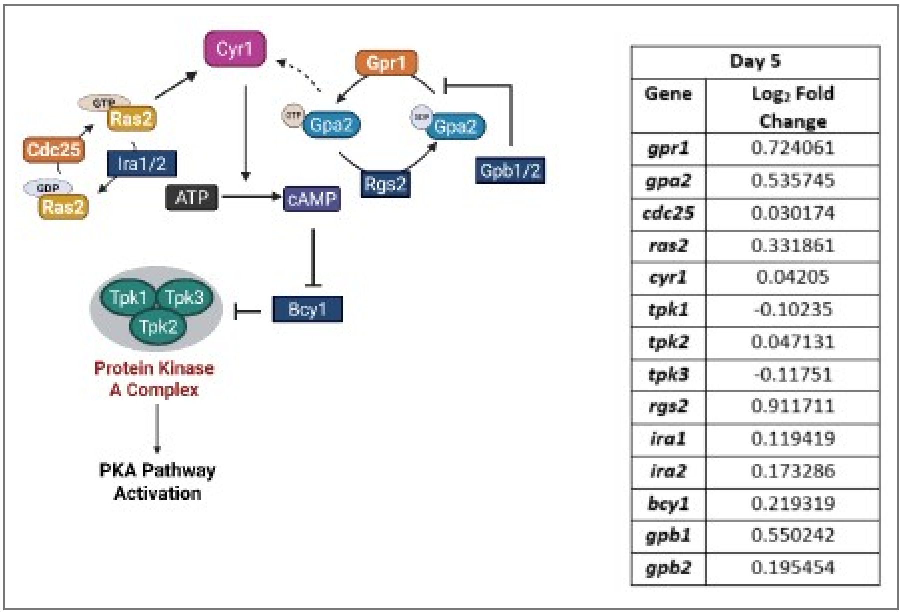
> *(6) The comparative RT-qPCR in Figure 5 did not account for sulfur metabolism genes, whereas it was focused only on virulence and hyphal differentiation. Is there data to support the levels of sulfur metabolism genes?*

We thank the reviewer for this feedback. We wish to respectfully clarify that the data pertaining to expression of sulfur metabolism genes in the presence of 2DG or in the ΔΔpfk1 strain in C. albicans are already included and discussed within the manuscript. These results can be found in Figure 4, panels D and I, respectively.

> *(7) To validate the proposed interlink between sulfur metabolism and virulence, it is recommended that the gene sets (illustrated in Figure 6) be consistently included across all comparative data included throughout the comparisons. Excluding sulfur metabolism genes in Figure 5 prevents the experiment from demonstrating the coordinated role of glycolysis perturbation → sulfur metabolism → virulence. The same is true for other comparisons, where the lack of data on Met30, Met4, etc., makes it hard.to connect the hypothesis. It is also recommended to check the gene expression of other genes related to the cAMP pathway and report them to confirm the cAMP-independent mechanism. For example, gap2 deletion was used to confirm the effects of cAMP supplementation, but the expression of this gene was not assessed in the RNA-seq analysis in Figure 2. It would be beneficial to show the expression of cAMP-related genes to completely confirm that they do not play a role in the claims in Figure 2.*

We thank the reviewer for this valuable feedback. The transcriptional analysis of the sulfur metabolism genes in the presence of 2DG and the ΔΔpfk1 strain is shown in Figures 4D and 4I.

Our RNA-seq analysis (Author response image 1) confirms that there is no significant transcriptional change in the expression of cAMP-PKA pathway associated genes (Log2 fold change ≥ 1 for upregulated genes and Log2 fold change ≤ -1 for downregulated genes) in 2DG treated cells compared to the untreated control cells, reinforcing our conclusion that the glycolytic regulation of fungal morphogenesis is mediated through a cAMP-PKA pathway independent mechanism.



Day 5	
Gene	Log_2 Fold Change
gpr1	0.724061
gpa2	0.535745
cdc25	0.030174
ras2	0.331861
cyr1	0.04205
tpk1	-0.10235
tpk2	0.047131
tpk3	-0.11751
rgs2	0.911711
ira1	0.119419
ira2	0.173286
bcy1	0.219319
gpb1	0.550242
gpb2	0.195454

Author response image 1.

> *(8) Although the NAC supplementation study is included in the new version of the article compared to the previous version in BioRxiv (May 2025), the link to sulfur metabolism is not well characterized in Figure 5 and their related datasets. The main focus of the manuscript is to delineate the role of sulfur metabolism; hence, it is anticipated that Figure 5 will include sulfur-related metabolic genes and their links to pfk1 deletion, using RT-PCR measurements as shown for the virulence genes.*

We thank the reviewer for this question. The relevant data are indeed present within the current submission. We respectfully direct the reviewer's attention to Figure 4, panels D and I, where the data pertaining to expression of sulfur metabolism genes in the presence of 2DG or in the ΔΔpfk1 strain in C. albicans can be found.

> *(9) The manuscript would benefit from more information added to the introduction section and literature supports for some of the findings reported earlier, including the role of (i) cAMP-PKA and MAPK pathways, (ii) what is known in the literature that reports about the treatment with 2DG (role of Snf1, HXT1, and HXT3), as well as how gpa2 is involved. Some sentences in the manuscripts are repetitive; it would be beneficial to add more relevant sections to the introduction and discussion to clarify the rationale for gene choices.*

We thank the reviewer for this valuable feedback. We have incorporated these changes in our revised manuscript.

> ***Recommendations for the authors:***
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> ***Reviewer #1 (Recommendations for the authors):***
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> *(1) Line 107: As morphological transitions are indeed a conserved phenomenon across fungal species, hosts & environmental niches, the authors could refer to a few more here (infection structures like appressoria; fruiting bodies, etc.).*

We thank the reviewer for this valuable feedback. We have incorporated these changes in our revised manuscript.

> *Line 119/120: That's a bit misleading in my opinion. Gpr1 acts as a key sensor of external carbon, while Ras proteins control the cAMP pathway as intracellular sensory proteins. That should be stated more clearly. cAMP is the output and not the sensor.*

We appreciate the reviewer's detailed attention to this signaling network. We have revised the manuscript to precisely reflect this established signaling hierarchy for maximum clarity.

> *(2) Line 180: ..differentiation*

We thank the reviewer for this valuable feedback. We have incorporated this change in our revised manuscript.

> *(3) Figure 1 panels C & F. The authors should provide the same scale for all experiments. Otherwise, the interpretation can be difficult. The same applies to the different bar plots in Figure 4. Have the authors quantified pseudohyphal differentiation in the cAMP add-back assays? I agree that the chosen images look convincing, but they don't reflect quantitative analyses.*

We thank the reviewer for detailed and constructive feedback. We have changed the Y-axis and made it more uniform to improve the clarity of our data presentation in the revised manuscript.

We have also incorporated the quantitative analysis of the cAMP add-back assays in S. cerevisiae, in Figure 2 Panel L.

> *(4) Line 367/68: "cysteine or methionine was able to completely rescue". Here, the authors should phrase their wording more carefully. Figure 3C shows the complete rescue of the phenotype qualitatively, but Figure 3D clearly shows that there are differences between the supplementation of cysteine and methionine, with the latter not fully restoring the phenotype.*

We sincerely appreciate the reviewer's meticulous attention to the data interpretation. We fully agree that the initial phrasing in lines 367/368 requires adjustment, as Figure 3D establishes a quantitative difference in the efficiency of phenotypic rescue between cysteine and methionine supplementation. We have revised the text to articulate this difference.

> *(5) Line 568: Here, apparently, the ability to rescue the differentiation phenotype is reversed compared to the experiment with S. cerevisiae. Cysteine only results in ~20% hyphal cells, while methionine restores to wild-type-like hyphal formation. Can the authors comment on where these differences might originate from? Is there a difference in the uptake of cysteine vs. methionine in the two species or consumption rates?*

We thank the reviewer for their detailed and constructive feedback. We believe this phenotypic difference can be due to the distinct metabolic prioritization of sulfur amino acids in C. albicans. Methionine is a known trigger for hyphal differentiation in C. albicans and serves as the immediate precursor for the universal methyl donor, S-adenosylmethionine (SAM) (Schrevens et al., 2018). (Kraidlova et al., 2016). The morphological transition to hyphae involves a complex regulatory cascade which requires high rates of methylation, and this requires a rapid and direct conversion of methionine into SAM (Kraidlova et al., 2016; Schrevens et al., 2018). Cysteine, however, must first be converted into methionine via the transsulfuration pathway to produce SAM, making it metabolically less efficient for these aforesaid processes.

> ***Reviewer #2 (Recommendations for the authors):***

The study's comprehensive experimental approach with integrating pharmacological inhibition, genetic manipulation, transcriptomics, and infection animal model, provides strong evidence for a conserved mechanism, though some aspects need further clarification.

> *Major Comments:*
>
> *(1) While the data suggest that glycolysis affects Met30 activity post-translationally, the underlying mechanism remains speculative. The authors should perform co-immunoprecipitation or ubiquitination assays to confirm whether glycolytic perturbation alters Met30-SCF complex interactions or Met4 ubiquitination levels.*

We thank the reviewer for this valuable feedback. The activity of the SCF^{Met30} E3 ubiquitin ligase, mediated by the F box protein Met30, is dynamically regulated through both proteolytic degradation and its dissociation from the SCF complex, to coordinate sulfur metabolism and cell cycle progression (Smothers et al., 2000; Yen et al., 2005). Our transcriptomic (RNA-seq analysis) and protein expression analysis (Fig. 3J) confirms that Met30 expression is not differentially regulated in the presence of 2DG, effectively eliminating changes in synthesis or SCF^{Met30} proteasomal degradation as the dominant regulatory mechanism. This observation is consistent with the established paradigm wherein stress signals, such as cadmium (Cd^{2+}) exposure, rapidly inactivates the SCF^{Met30} E3 ubiquitin ligase via the dissociation of Met30 from the Skp1 subunit of the SCF complex (Lauinger et al., 2024; Yen et al., 2005). We therefore propose that active glycolytic flux modulates SCF^{Met30} activity post-translationally, specifically by triggering Met30 detachment from the SCF complex. This mechanism would stabilize the primary substrate, the transcription factor Met4, thus promoting the biosynthesis of sulfur-containing amino acids. Mechanistic validation of this hypothesis, particularly the assessment of Met30 dissociation from the SCF^{Met30} complex via immunoprecipitation (IP), is technically challenging. Since these experiments will involve isolation of cells from colonies undergoing pseudohyphal differentiation, on solid media (given that pseudohyphal differentiation does not occur in liquid media that is limiting for nitrogen (Gancedo, 2001; Gimeno et al., 1992)), current cell yields (OD^{600}≈1 from ≈80-100 colonies) are significantly below the amount of cells that is needed to obtain the required amount of total protein concentration, for standard pull down assays (OD600≈600-800 is required to achieve 1-2 mg/ml of total protein which is the standard requirement for pull down protocols in S. cerevisiae (Lauinger et al., 2024)).

Given that the primary objective of our study is to establish the novel regulatory link between glycolysis and sulfur metabolism in the context of fungal morphogenesis, we would like to explore these crucial mechanistic details, in depth, in a subsequent study.

> *(2) 2DG can exert pleiotropic effects unrelated to glycolytic inhibition (e.g., ER stress, autophagy induction). The authors are encouraged to perform complementary metabolic flux analyses, such as quantification of glycolytic intermediates or ATP levels, to confirm specific glycolytic inhibition.*

We appreciate the reviewer's concern regarding the potential pleiotropic effects of 2DG. While we acknowledge that 2DG may induce secondary cellular stress, we are confident that the observed phenotypes are robustly attributed to glycolytic inhibition based on our complementary genetic evidence. Specifically, the deletion strains ΔΔpfk1 and ΔΔadh1, which genetically perturb distinct steps in glycolysis, recapitulate the phenotypic results observed with 2DG treatment. Given this strong congruence between chemical inhibition and specific genetic deletions of key glycolytic enzymes, we are confident that our observed phenotypes are predominantly driven by the perturbation of the glycolytic pathway by 2DG.

> *(3) The differential rescue effects (cysteine-only in inhibitor assays vs. both cysteine and methionine in genetic mutants) require further explanation. The authors should discuss potential differences in metabolic interconversion or amino acid transport that may account for this observation.*

We thank the reviewer for their valuable feedback. One explanation for the observed differential rescue effects of cysteine and methionine can be due to the distinct amino acid transport systems used by S. cerevisiae to transport these amino acids. S. cerevisiae primarily uses multiple, lowaffinity permeases (Gap1, Bap2, Bap3, Tat1, Tat2, Agp1, Gnp1, Yct1) for cysteine transport, while relying on a limited set of high-affinity transporters (like Mup1) for methionine transport, with the added complexity that its methionine transporters can also transport cysteine (Düring-Olsen et al., 1999; Huang et al., 2017; Kosugi et al., 2001; Menant et al., 2006). Hence, it is likely that cysteine uptake could be happening at a higher efficiency in S. cerevisiae compared to methionine uptake. Therefore, to achieve a comparable functional rescue by exogenous supplementation of methionine, it is necessary to use a higher concentration of methionine. When we performed our rescue experiments using higher concentrations of methionine, we did not see any rescue of pseudohyphal differentiation in the presence of 2DG and in fact we noticed that, at higher concentrations of methionine, the wild-type strain failed to undergo pseudohyphal differentiation even in the absence of 2DG. This is likely due to the fact that increasing the methionine concentration raises the overall nitrogen content of the medium, thereby making the medium less nitrogen-starved. This presents a major experimental constraint, as pseudohyphal differentiation is strictly dependent on nitrogen limitation, and the elevated nitrogen resulting from the higher methionine concentration can inhibit pseudohyphal differentiation.

> *(4) NAC may influence host redox balance or immune responses. The discussion should consider whether the observed virulence rescue could partly result from host-directed effects.*

We thank the reviewer for this valuable feedback. We acknowledge the role of NAC in host directed immune response. It is important to note that, in the context of certain bacterial pathogens, NAC has been reported to augment cellular respiration, subsequently increasing Reactive Oxygen Species (ROS) generation, which contributes to pathogen clearance (Shee et al., 2022). Interestingly, in our study, NAC supplementation to the mice was given prior to the infection and maintained continuously throughout the duration of the experiment. This continuous supply of NAC likely contributes to the rescue of virulence defects exhibited by the ΔΔpfk1 strain (Fig. 5I and J). Essentially, NAC likely allows the mutant to fully activate its essential virulence strategies (including morphological switching), to cause a successful infection in the host. As per the reviewer suggestion, this has been included in the discussion section of the manuscript.

> ***Reviewer #3 (Recommendations for the authors):***
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> *Most of the comments related to improving the manuscript have been provided in the public review. Here are some specifics for the authors to consider:*
>
> *(1) It is important to clarify the rationale for choosing specific gene deletions over other key genes (e.g., Met32 and Met30) and explain why Met4 was not included, given its proposed central role in Figure 6.*

We sincerely thank the reviewer for this insightful query regarding our selection of the met32 for our gene deletion experiments. The choice of ΔΔmet32 strain was strategically motivated by its unique phenotypic properties within the de novo biosynthesis of sulfur-containing amino acids pathway. While deletions of most the genes that encode for proteins involved in the de novo biosynthesis of sulfurcontaining amino acids, result in auxotrophy for methionine or cysteine, ΔΔmet32 strain does not exhibit this phenotype (Blaiseau et al., 1997). This key distinction is attributed to the functional redundancy provided by the paralogous gene, met31 (Blaiseau et al., 1997). Crucially, given that the deletion of the central transcriptional regulator, met4, results in cysteine/methionine auxotrophy, the use of the

ΔΔmet32 strain provides an essential, viable experimental model for investigating the role of sulfur metabolism during pseudohyphal differentiation in S. cerevisiae.

> *(2) Comparison of consistent gene and protein expression data (Met30, Met4, Met32) across all relevant figures and analyses would strengthen the mechanistic connection in a better way. Some data that might help connect the sections is not included; please see the public review for more details.*

We thank the reviewer for this valuable input, which helps us to clarify the scope of our transcriptomic analysis. Our decision to focus our RT-qPCR experiments on downstream targets, while excluding Met4 and Met30 from the RT-qPCR analysis, is based on their known regulatory mechanisms. Met4 activity is predominantly regulated by post-translational ubiquitination by the SCFMet30 complex followed by its degradation (Rouillon et al., 2000; Shrivastava et al., 2021; Smothers et al., 2000) while Met30 activity is primarily regulated by its auto-degradation or its dissociation from the SCFMet30 complex (Lauinger et al., 2024; Smothers et al., 2000; Yen et al., 2005). Consistent with this, our RNA-Seq results indicate that neither met4 nor met30 transcripts are differentially expressed, in response to 2DG addition. For all our RT-qPCR analysis in S. cerevisiae and C. albicans, we have consistently used the same set of sulfur metabolism genes and these include met32, met3, met5, met10 and met17. Our data on protein expression analysis of Met30 in S, cerevisiae (Fig. 3J) confirms that Met30 expression is not differentially regulated in the presence of 2DG, effectively eliminating changes in synthesis or SCFMet30 proteasomal degradation as the dominant regulatory mechanism.

> *(3) Suggested to include metabolomic profiling (cysteine, methionine, and intermediate metabolites) to substantiate the proposed metabolic flux between glycolysis and sulfur metabolism.*

We thank the reviewer for this valuable input. Our pseudohyphal/hyphal differentiation assays show that the defects induced by glycolytic perturbation is fully rescued by the exogenous supplementation of sulfur-containing amino acids, cysteine or methionine. Since these data conclusively demonstrate that the primary metabolic limitation caused by the perturbation of glycolysis, which leads to filamentation defects, is sulfur metabolism, we posit that performing comprehensive metabolic profiling would primarily reconfirm the aforesaid results. We believe that our in vitro and in vivo sulfur add-back experiments sufficiently substantiate the novel regulatory metabolic link between glycolysis and sulfur-metabolism.

> *(4) Data on the effects of Met30 deletion on cell growth are currently not included, and relevant controls should be included to ensure observed phenotypes are not due to general growth defects.*

We are grateful to the reviewer for this constructive feedback. To address the potential impact of met30 deletion on cell growth, we have included new data (Suppl. Fig. 4A) demonstrating that the deletion of a single copy of met30 in diploid S. cerevisiae does not compromise overall growth under nitrogen-limiting conditions as the Δmet30 strain grows similar to the wild-type strain.

> *(5) Expanding RT-qPCR and data from transcriptomic analyses to include sulfur metabolism genes and key cAMP pathway genes to confirm the proposed cAMP-independent mechanism during virulence characterization is necessary.*

We thank the reviewer for this valuable feedback. The transcriptional analysis of the sulfur metabolism genes in the presence of 2DG and the ΔΔpfk1 strain is shown in Figures 4D and 4I.

In order to confirm that glycolysis is critical for fungal morphogenesis in a cAMP-PKA pathway independent manner under nitrogen-limiting conditions in C. albicans, we performed cAMP add-back assays. Interestingly, corroborating our S. cerevisiae data, the exogenous addition of cAMP failed to rescue hyphal differentiation defect caused by the perturbation of glycolysis through 2DG addition or by the deletion of the pfk1 gene, under nitrogen-limiting condition in C. albicans. This data is now included in Suppl. Fig. 5B.

> *(6) Enhancing the introduction and discussion by providing a clearer rationale for gene selection and more detailed references to established pathways (cAMP-PKA, MAPK, Snf1/HXT regulation, gpa2 involvement) is needed to reinstate the hypothesis.*

We thank the reviewer for this valuable feedback. We have incorporated these changes in our revised manuscript.

> *(7) Reducing redundancy in the text and improving figure consistency, particularly by ensuring that the gene sets depicted in Figure 6 are represented across all datasets, would strengthen the interconnections among sections.*

We thank the reviewer for this valuable feedback. We have incorporated these changes in our revised manuscript.

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